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# Laser assisted surface activation and selective copper metallization for ceramic substrates: a review

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## Abstract

Ceramic materials, with high thermal conductivity, low dielectric constant and excellent strength, exhibited outstanding heat dissipation and stability in electronic application scenarios, but the regulation of ceramic conductivity has become a huge challenge in their engineering promotion. Therefore, exploring the improvement of ceramic physicochemical properties and surface metallization technology has become an important engine for the leapfrog development of electronic ceramic devices. Among the existing surface treatment methods, laser surface activation and metallization have been paid attention to because of their advantages of precision, high efficiency, environmental protection and easy industrialization, so this review also aims to conduct a

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comprehensive investigation and in-depth discussion of this technology. First, the principle of laser ablation and its effect on the physical and chemical properties of ceramic materials are discussed in detail. Next, in the process of electroless copper plating (ECP), copper ions fully adhere to the laser structured ceramic surface and form an interlock at the bonding interface, which will help to improve the bonding strength of the metallized coating. Then, the ceramic substrate after laser ablation may be prone to corrosion risk, but the microstructure formed on the laser-activated surface plays a crucial role in the preparation of high-quality metal coatings. Finally, future development directions, including the innovation of ceramic metallization and the invention of advanced electronic devices, are prospected.

Keywords

Ceramic substrate; Laser activation; Surface microstructure; Selective metallization; Electroless copper plating; Bonding strength

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1. Introduction

In recent years, ceramic materials have gained significant popularity in the electronics industry due to their superior electrical and mechanical properties [1], [2], [3], and typical applications include hybrid integrated circuits, optoelectronic device substrates and microfluidic circuit boards [4], as illustrated in Fig. 1. The standard electronic substrate for these devices consists of ceramic substrates (usually alumina and aluminum nitride) and circuit layers (usually copper) [5,6], among which the adhesion of copper to the ceramic substrate is a key issue affecting the performance and reliability of system [7]. Effectively controlling the performance of the prepared conductive regions poses a challenge due to the thermal nature of the ceramic decomposition process and the

Declaration of competing interest

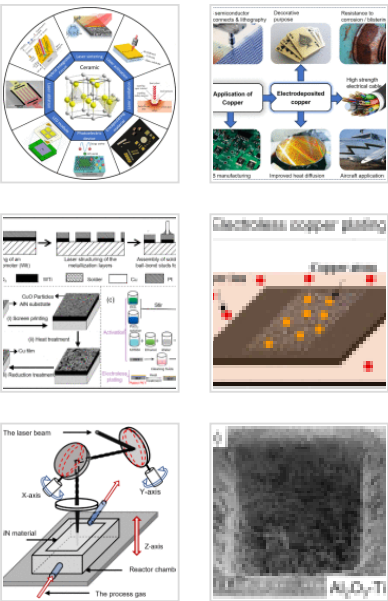
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Figures (26)

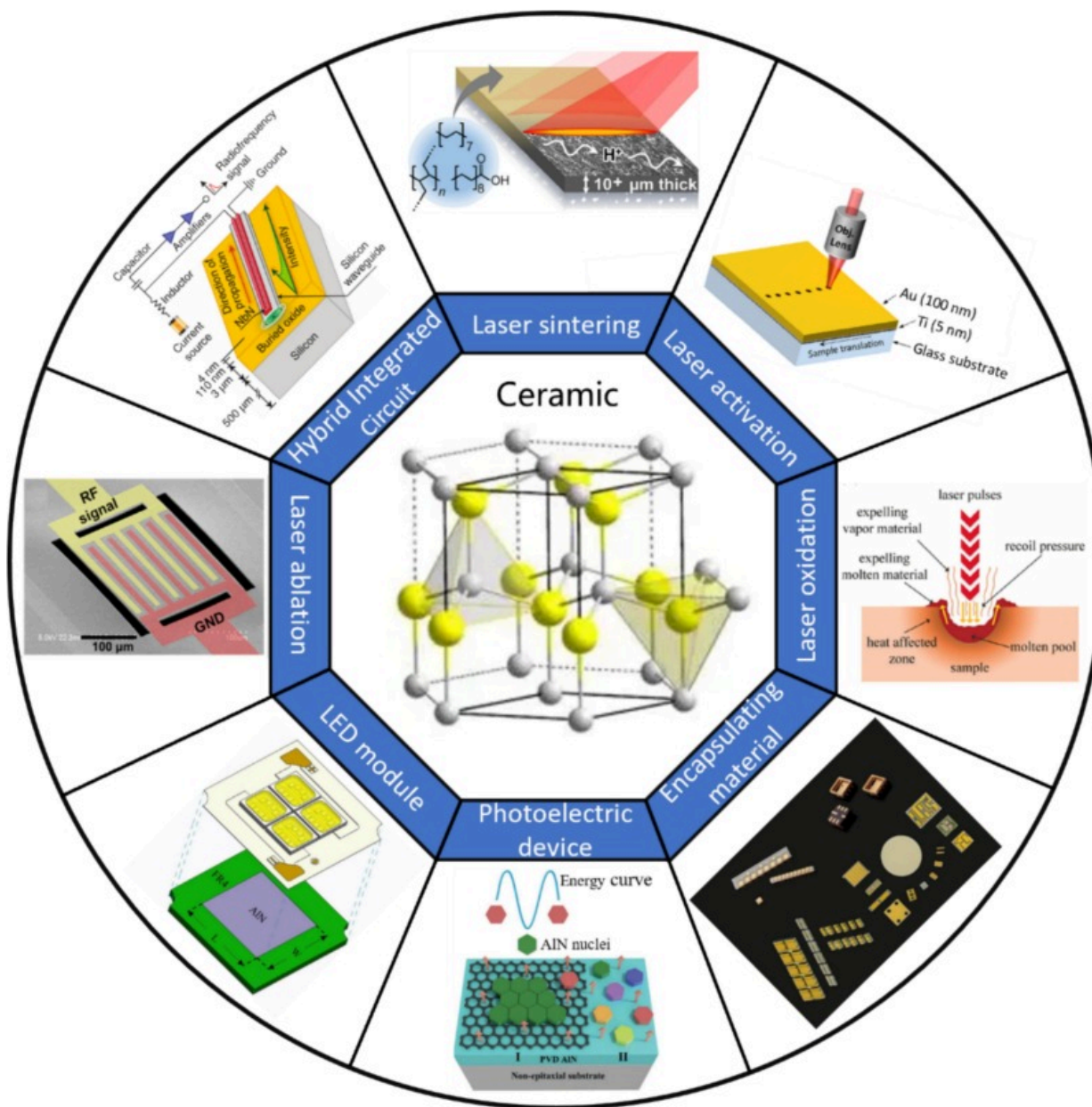


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material's high thermal conductivity. ECP is a well-established method for depositing metallic coatings onto ceramic surfaces through a catalytic redox reaction without requiring an external electrical current [8]. Owing to its advantages of low processing temperature, high deposition rate, and low cost, this technique is increasingly utilized in industries such as semiconductors and aerospace [9,10].



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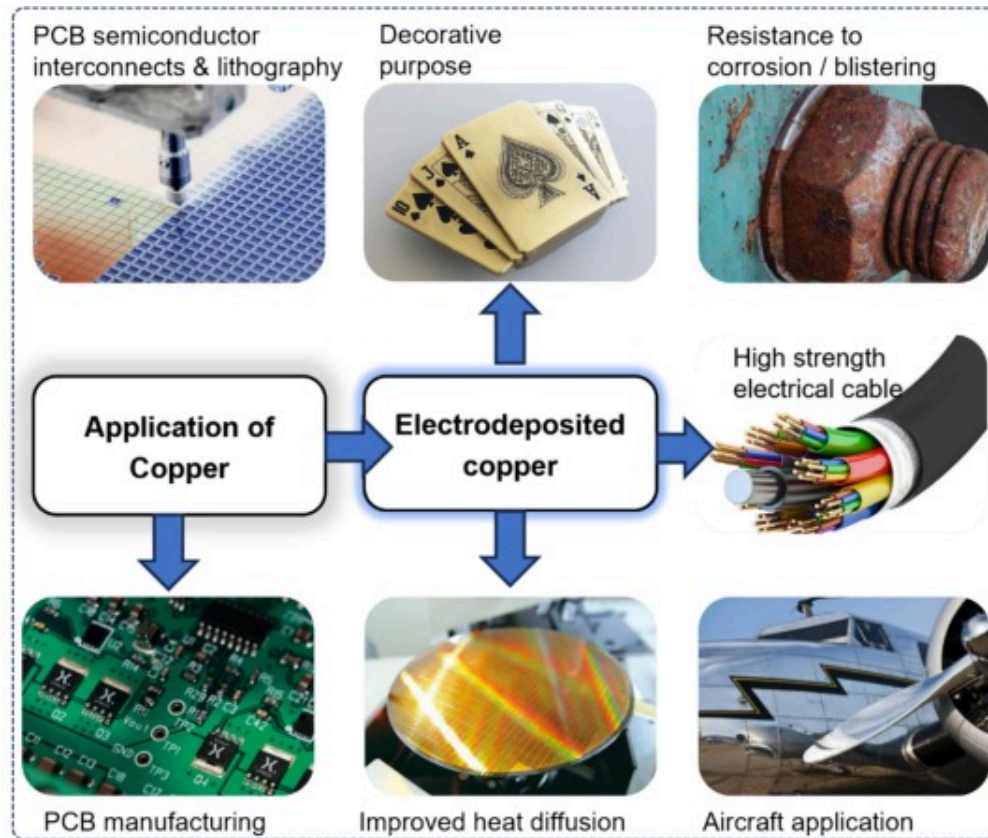
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Fig. 1. Application of laser processing technology in ceramics [[11], [12], [13], [14], [15]].

ECP has emerged as a significant trend in the manufacturing of PCB circuit boards. It is utilized to form conductive lines and achieve hole metallization. Furthermore, the technology finds broad application in decorative finishes, aerospace components, and other fields due to its enhanced properties, including thermal dissipation, corrosion resistance, and anti-foaming characteristics, as shown in [Fig. 2](#). Although ECP has many advantages, the weak adhesion generated by the van der Waals force between the metal layer atoms and the ceramic substrate hinders the expansion of this technology [[16,17](#)], and the properties and structure of the ceramics ultimately determine its performance. Ma et al. [[18](#)] reported that a flat and dense nano-composite coating was prepared by activating the ceramic substrate with nano-gold. Although this method simplifies the ceramic surface pretreatment process, the low adhesion of the coating surface limits the practical application. Antony et al. [[19](#)] fabricated microstructures on glass substrates using laser technology, subsequently depositing copper layers via magnetron sputtering and applying functional layers through electroplating. This approach produced copper films with a thickness of 8  $\mu\text{m}$ , which have excellent adhesion strength and low surface roughness. However, the entire technological process reveals the inherent complexity of the above-mentioned method. Raumel et al. [[20](#)] developed a novel direct metallization process for BN-AlN composite ceramics utilizing laser direct structuring. The laser-treated ceramic surface exhibits significantly more complex microscale surface profiles and porous structures, thereby enhancing mechanical interlocking (or riveting effect) and bond strength between the copper layer and ceramic substrate. Specifically, the adhesion between metal layers and ceramic substrates is typically governed by chemically or physically modified surfaces treatments. Most research has focused on employing diverse pretreatment steps to enhance bond strength. Various thermal, mechanical, chemical, and hybrid treatments have been developed to modify the surface properties of ceramic materials. Popular ceramic metallization techniques include direct copper plating [[21](#)], thin-film metallization [[22](#)], thick-film metallization [[23](#)], and laser-induced selective metallization [[24](#)]. Selective laser ablation of AlN ceramic substrates is a promising technique that integrates laser processing with electroless plating to deposit conductive metallic patterns on material surfaces [[25,26](#)]. Compared to other ceramic surface



treatment methods, this technique improves electrical conductivity in the ablated regions, ensures robust adhesion between the ceramic substrate and metal coating, and generates no toxic byproducts. Furthermore, it simplifies the manufacturing workflow and significantly reduces costs relative to conventional dielectric metallization processing.



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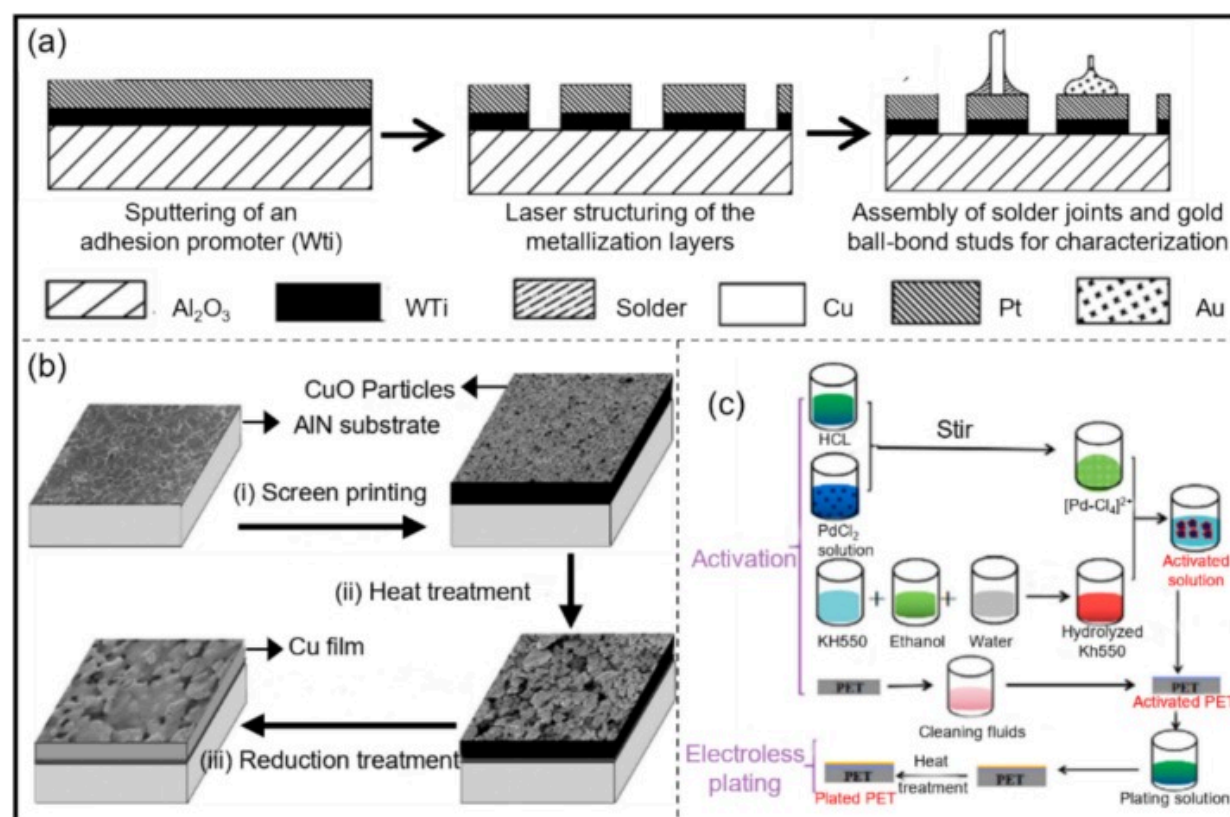
Fig. 2. Application of chemical deposition for copper thin-film fabrication.

AlN ceramics used in radiofrequency and electronic devices require metallization to establish a reliable connection between ceramic and metal components. This integration

forms metal-ceramic structures where the metal acts as electrodes, enabling electrical current conduction [27]. To achieve strong copper-substrate adhesion, direct copper plating can be applied to ceramic surfaces, facilitating the direct fabrication of high-power devices. As demonstrated by Squiers et al. [28], pure copper foils directly bonded to ceramic substrates effectively meet power electronics requirements – including low power consumption, ease of fabrication, thick metallization, and high current capacity – owing to their excellent current-carrying capacity and thermal conductivity. In the study by Mei et al. [29], copper residues were detected on ceramic surfaces following direct copper plating, and the residual copper deposits can induce electrolytic metal migration [30], where dendritic crystals bridge adjacent conductors on the insulating substrate, resulting in electrical short circuits [31,32]. This failure mechanism is frequently observed in microelectronic devices and has a significant impact on the reliability of direct bonded copper technology within power electronics applications.

Thin-film metallization is also a widely employed technique in ceramic metallization treatments. This method adheres metal films onto the surface of ceramic substrates via vacuum coating technology. Since vapor deposition can, in principle, deposit various metals onto any ceramic substrate, the resulting metal layers exhibit both high uniformity and strong bonding strength [33,34]. Wu et al. [35] prepared composite films on ceramic substrates by using magnetron sputtering. The results proved that the mutual diffusion of Ti/Cu mixed metal structures led to a reduction in shear strength. Ti/Ni/Au is a relatively ideal composite film structure in the metallization of ceramic substrate films, and its permeability is much higher than that of Ni films. In Fig. 3 (a), Kiele et al. [36] sputtered a 43nm WTi layer as an adhesion promoter on the  $\text{Al}_2\text{O}_3$  substrate, then yielding a 500nm thin-film of Pt with an adhesion strength of  $32.75 \pm 7.08 \text{ MPa}$ . Immediately after, laser structuring of metallization layers was carried out to assemble solder joints and gold ball bonds for subsequent characterization. The metallization of sputtered WTi-Pt exhibited excellent long-term stability in both high-temperature air and saline solution. However, thin film metallization requires the surface of the ceramic substrate have extremely high purity and completely remove adsorbed water vapor and other substances, and it also

needs a subsequent pattern-making process to prepare metal wire patterns, which are all relatively expensive.



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Fig. 3. Metallization method: (a) thin film metallization [36]; (b) thick film metallization [23]; (c) chemical metallization [41].

Thick film metallization is a method of directly depositing conductive paste on substrate through micro-flow direct writing technology such as screen-printing [37], micro-straight writing [38] and inkjet printing, and sintering at high temperature to form conductive lines and electrodes. Hlina and Reboun [39] found that the adhesion mechanism of AlN is based on the  $\text{Cu}_2\text{O}$  layer formed on the surface, while the adhesion mechanism of  $\text{Al}_2\text{O}_3$  is



provided by  $\text{SiO}_2$  and  $\text{Bi}_2\text{O}_3$  glass frits with high permeability. Since there is no glass phase penetrates into the  $\text{AlN}$  substrate, the adhesion mechanism of  $\text{AlN}$  is the same as that of  $\text{Al}_2\text{O}_3$  substrate. Compared with  $\text{Al}_2\text{O}_3$  substrate, copper metallized  $\text{AlN}$  has stronger thermal and shock resistance characteristics, and can be used in complex interconnected and multilayer circuits [40]. Zhang et al. [23] coated  $\text{CuO}$  electronic paste on  $\text{AlN}$  substrate through screen printing, and heat-treated in air atmosphere at  $1050^\circ\text{C}$ ,  $1075^\circ\text{C}$  and  $1100^\circ\text{C}$  for 30min to form an interface layer, and then reduced in  $80\%\text{N}_2+20\%\text{H}_2$  atmosphere at  $500^\circ\text{C}$ , as illustrated in Fig. 3 (b). The results show that the optimum bond strength can be obtained at  $1075^\circ\text{C}$ , because  $\text{CuAlO}_2$  and  $\text{CuAl}_2\text{O}_4$  are formed in the interface layer, which can act as an intermediate phase to reduce the residual stress and mechanical interlock at the interface of  $\text{Cu}$  film and  $\text{AlN}$ , and obtain a higher adhesion strength through chemical and physical interactions. The thick-film metallization process is straightforward, however, due to the size of conductive paste particles and the wire mesh, performance of the resulting conductive circuit is comparatively limited.

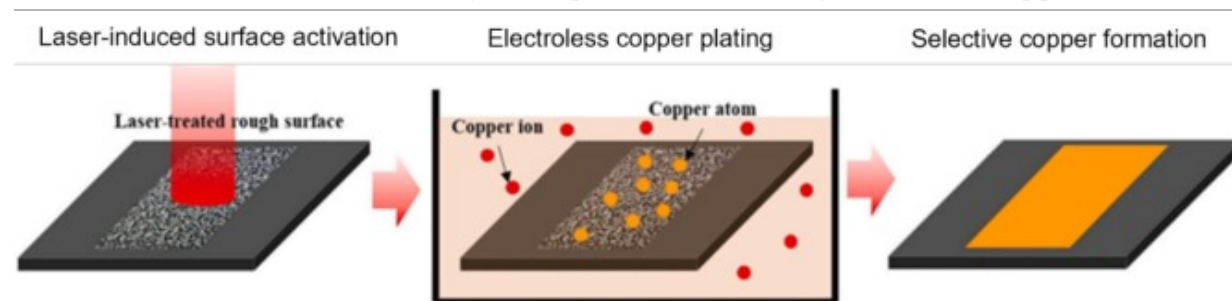
The initial process of chemical metallization involves cleaning and roughening the ceramic surface to remove organic contaminants and enlarge its surface area. A reducing agent and noble-metal salts are then adsorbed onto the substrate, forming active nanoscale metal nuclei that act as catalytic centers for subsequent metal deposition. Huang et al. [41] developed a novel catalyst solution composed of palladium nanoparticles/polymer brushes ( $\text{Pd/PB}$ ). A  $\text{PdCl}_2$  solution was formulated using deionized water, ethanol and 3-aminopropyltriethoxysilane ( $\text{KH550}$ ) for the simultaneous modification and activation of polyethylene terephthalate ( $\text{PET}$ ) substrates. The  $\text{Pd}$  nanoparticles were adsorbed along with  $-\text{NH}_2$  groups by the  $\text{PB}$ , while the silane coupling agent-based solution formed an active layer with a  $\text{Pd/PB}$  structure on the surface of  $\text{PET}$ , which eventually presents an  $\text{Ag/Pd/PB}$  structure and can be served as a seed for nickel plating of catalytic electrodes. As illustrated in Fig. 3 (c), this approach provided a streamlined electroless plating route, yielding a coating with a maximum effective thickness of  $0.8\mu\text{m}$ , a fracture energy of  $0.4\text{J/m}^2$ , and adhesion strength graded 5B.

Although chemical metallization enables the fabrication of high-resolution fine circuits at low temperatures, the resulting coatings are often characterized by internal stress and relatively low bond strength.

The results show that the optimum bond strength can be obtained at 1075°C, because  $\text{CuAlO}_2$  and  $\text{CuAl}_2\text{O}_4$  are formed in the interface layer, which can act as an intermediate phase to reduce the residual stress and mechanical interlock at the interface of Cu film and AlN, and obtain a higher adhesion strength through chemical and physical interactions. The thick-film metallization process is straightforward, however, due to the size of conductive paste particles and the wire mesh, performance of the resulting conductive circuit is comparatively limited.

With the development of laser technology, laser ablation has been increasingly employed as surface pretreatment prior to bonding [42,43], owing to its advantages including high precision, reduced circuit board size and optimized circuit architecture, which is particularly suitable for the fabrication of microfluidic circuits [44]. This technique enables the surface of materials to be processed into micro and nano structures while changing their chemical composition [45]. Compared with other traditional surface treatments, laser processing technology has the advantages of low energy consumption, no pollution, etc. At the same time, it can be applied to most materials and the micro and nano structures processed can be customized in design and manufacturing. To obtain conductive layers with low resistance and stable chemical structure, Ntonczak et al. [46] conducted a series of study on laser ablation parameters and protective gases for laser-induced formation of conductive structures on ceramic substrate surfaces, demonstrating that direct metallization of AlN ceramics can be applied to metamaterial prototype in the microwave and terahertz ranges. Sartinska et al. [47] employed nanosecond laser processing under a nitrogen atmosphere to treat the surfaces of various materials and investigated the post-ablation surface structures. A novel method for fabricating nanowires and nanotubes on material surfaces was proposed to obtain a uniform nanoscale grain structure. Han et al. [48] combined laser-induced surface modification with ECP to achieve selective metallization of piezoelectric ceramics and to enhance their

adhesion properties. The results demonstrate that surfaces treated with laser ablation at a scanning speed of 200mm/s exhibit minimal laser-induced damage, superior adhesion strength, and improved aging resistance, as illustrated in Fig. 4. Sun et al. [49] developed an innovative approach for fabricating low-resistance conductive layers on AlN ceramics in ambient air, thereby bypassing conventional controlled-atmosphere requirements. For a circuit measuring 10mm×0.5mm, the electrical resistance was reduced to 0.26Ω. Importantly, this technique not only streamlines the metallization process for AlN ceramics but also meets the stringent requirements of integrated circuit applications.



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Fig. 4. Schematic diagram of selective metallization of piezoelectric ceramics by laser induced surface modification combined with ECP [48].

Although certain metallization techniques provide strong bonding strength, the durability of the metallized surfaces remains relatively poor when exposed to air or corrosive environments. Table 1 presents a comparison of various ceramic metallization methods. Compared to previous several technologies, laser-induced selective metallization offers greater flexibility in adjusting ablation conditions and parameters [50]. As a non-contact processing method, laser does not cause mechanical extrusion or mechanical stress to the material, and has a wide range of processing capabilities without pollution. A large number of studies have shown that laser ablation treatment can effectively improve the surface properties of ceramic materials [51] and the bonding performance between ceramic substrates and metal films. However, the interaction mechanism may vary

depending on the type of laser and ceramic material, and further in-depth research is still needed. The mechanism analysis of regulating the ceramic surface properties after laser ablation and the improvement of copper bonding properties after ECP will be discussed in detail in the following chapters. This paper reviews recent advances in laser processing of ceramics and subsequent ECP on their surfaces.

Table 1. Comparison of ceramic metallization methods.

| Method                   | Metal    | Adhesion strength<br>(kg/mm <sup>2</sup> ) | Merits and demerits                                                                         |                                                                                            |
|--------------------------|----------|--------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Thick film metallization | Cu       | 2.5                                        | High conductivity, but low bonding strength, not resistant to oxidation                     |                                                                                            |
|                          | Ag alloy | 2.8                                        | High bonding strength, oxidation resistance, but Ag is easy to diffuse                      |                                                                                            |
|                          | Au alloy | 2.7                                        | High bonding strength, oxidation resistance, good weldability, low diffusion, but expensive |                                                                                            |
|                          | W        | 3.6                                        | High bonding strength, poor weldability, no oxidation resistance                            |                                                                                            |
|                          | Mo-Mn    | 3.5                                        |                                                                                             |                                                                                            |
| Thin film metallization  | Ti-Pt-Au | 2.8                                        | Oxidation intolerance                                                                       | The film has high strength and can make fine wiring, but the equipment investment is large |
|                          | Ti-Ni-Au | 2.7                                        | Good brazing                                                                                |                                                                                            |
| Electroless plating      | Cu       | 2.4                                        | Low thermal resistance, erosion substrate, low                                              | Oxidation intolerance                                                                      |
|                          | Ni       | 2.6                                        | high temperature strength                                                                   | Low intensity                                                                              |

| Method                                                     | Metal | Adhesion strength<br>(kg/mm <sup>2</sup> ) | Merits and demerits                                                                                                                                                                                           |
|------------------------------------------------------------|-------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metallization<br>Direct copper<br>method<br>pouring method | Cu    | 1.6                                        | Not resistant to oxidation, low strength, difficult to refine graphics                                                                                                                                        |
| laser-induced<br>selective<br>metallization                |       |                                            | Low cost, high efficiency, simple equipment maintenance, agglomerates are generated on the surface of the metallized layer and are porous, the metallized layer has poor adhesion and uneven metal thickness. |

In this paper, the research progress of laser processing ceramics and ECP on the surface is reviewed. The effects of laser processing parameters on the thickness, bonding strength and resistance of ceramic substrate and copper layer are summarized. There are three main sections: [Section 2](#) reviews the principle and effect of laser processing ceramic surface; [Section 3](#) presents the main changes of ceramic surface properties (morphology, chemical composition and electrical conductivity) caused by various laser parameters are introduced. [Section 4](#) discusses the effect and failure analysis of laser ablation on bonding strength and conductivity of ECP on ceramic.

## 2. Laser processing of surface microstructure of ceramic materials

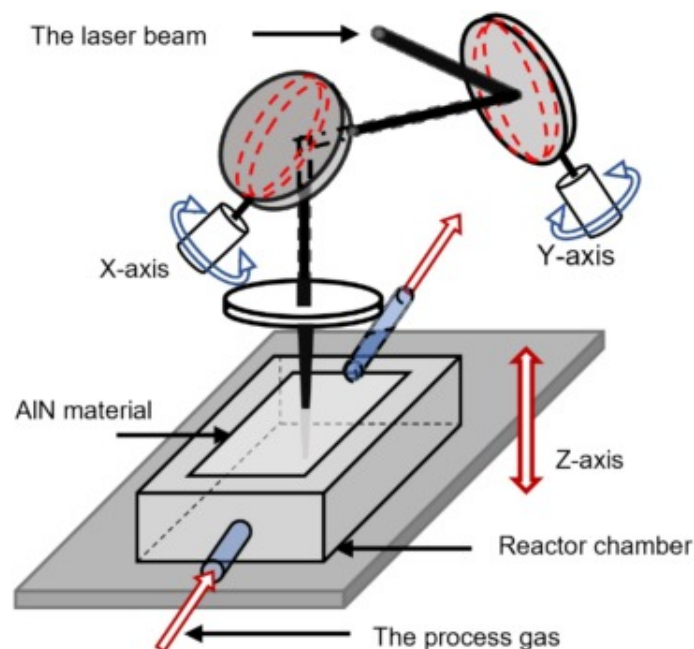
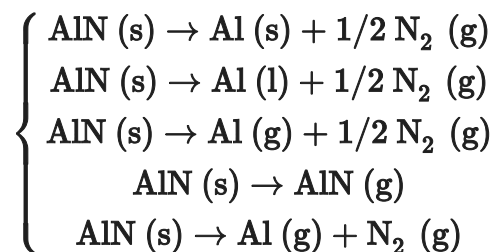
### 2.1. Principle analysis of laser processing ceramic surface microstructure

#### 2.1.1. Chemical reaction

AlN has remarkable advantages such as ultra-high band gap of 6.2eV, high thermal conductivity of  $<210 \text{ W/m}\cdot\text{K}$ , high resistivity of  $>10^{12} \Omega \text{ m}$ , and low coefficient of thermal expansion of  $\sim 5 \times 10^{-6} \text{ 1/K}$ , and is widely used in the fields of microelectronics and



optoelectronics. In particular, it has the possibility of direct metallization through thermal decomposition. The schematic system diagram of the AlN ceramic laser-induced selective metallization apparatus is presented in Fig. 5. Under laser beam irradiation with varying process parameters, the material surface undergoes distinct modifications. When transferred from the emission source to the ceramic surface, the laser energy can induce successive melting, vaporization, and solidification reactions [52]. In detail, the decomposition principle of AlN ceramics have been discussed in the following reaction equations (1a-e) [53,54].



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Fig. 5. System diagram of AlN ceramic laser processing [55].

Where g, l and s represent gas, liquid and solid, respectively. The enthalpy of reaction is denoted as  $H$ . Previous research indicated that the decomposed metallic aluminum in AlN transitions from solid phase to gas phase through the reactions (1a) to (1c) as the temperature increases under ambient pressure. At higher energy inputs, AlN either sublimates directly or decomposes into Al and nitrogen vapors according to reactions (1d) to (1e).

Consequently, achieving high-quality metallic Al layers requires optimal energy density selection. This parameter control enables solid Al layer formation on AlN ceramic surfaces while liberating nitrogen atoms as gaseous species. During high-power-density laser processing, localized temperatures rapidly exceed 2000K which allows the material to melt and vaporize directly. Under these conditions, AlN ceramics undergo two characteristic reactions: reacting with ambient oxygen and thermal decomposition into Al and diatomic nitrogen as described by Equations (2), (3), respectively.



### 2.1.2. Laser thermal effect and photochemical effect

The mechanism of laser machining ceramics can be divided into thermal and photochemical effects. In general, the condition that causes the removal of thermal effect is that the power density of the incident laser exceeds a certain threshold [53]. Its specific performance is that under laser irradiation, the surface of the material is heated, and when the surface dimension reaches the melting point, the material surface melts first. If the incident power density of the laser is increased (unit:  $\text{J}/\text{cm}^2$ ), so that the surface temperature rises to the boiling point, the surface of the material will begin to evaporate

to achieve the purpose of removal. When the irradiation intensity exceeds the threshold value, laser evaporation is caused by photochemical action [56]. Its specific performance is that the photon cuts off the internal molecular bonds, so that the material is in a loose state, causing the material to evaporate to achieve the purpose of removal. Therefore, the necessary condition for photochemical evaporation is that the energy of photons is greater than the binding energy of molecules. The interaction area between the ultraviolet laser and the material is mostly absorbed by the material, resulting in the decomposition of the material debris directly from the workpiece at a high speed, so the thermal effect is minimal, and the “cold processing” different from the ultrafast laser effect is achieved.

The chemical composition and preparation process have significant influence on the optical properties of AlN [57]. The refractive index and extinction coefficient of AlN ceramics with different compositions can be found in the reference [58]. According to Fresnel's formula, its reflectance  $R$  for each polarization direction is

$$R_{\lambda//} = \frac{(p - n_1 \sin \theta_1 \tan \theta_1)^2 + q^2}{(p + n_1 \sin \theta_1 \tan \theta_1)^2 + q^2} \times \frac{(n_1 \cos \theta_1 - p)^2}{(n_1 \cos \theta_1 + p)^2} \text{ for } p \text{ polarization} \quad (4)$$

$$R_{\lambda\perp} = \frac{(n_1 \cos \theta_1 - p)^2 + q^2}{(n_1 \cos \theta_1 + p)^2 + q^2} \text{ for } s \text{ polarization} \quad (5)$$

$$R_{\lambda c} = \frac{1}{2}(R_{//\lambda} + R_{\perp\lambda}) \text{ for } c \text{ polarization} \quad (6)$$

Where  $n_1$  is the refractive index in air and  $\theta_1$  is the incidence angle.  $p$  and  $q$  are:

$$[\sqrt{(n_\lambda^2 - k_\lambda^2 - n_1^2 \sin^2 \theta_1)^2 + 4n_\lambda^2 k_\lambda^2}] + (n_\lambda^2 - k_\lambda^2 - n_1^2 \sin^2 \theta_1) \quad (7)$$

$$p^2 = \frac{1}{2}[\sqrt{(n_\lambda^2 - k_\lambda^2 - n_1^2 \sin^2 \theta_1)^2 + 4n_\lambda^2 k_\lambda^2}] + (n_\lambda^2 - k_\lambda^2 - n_1^2 \sin^2 \theta_1) \quad (8)$$

### 2.1.3. Plasma plume characteristics

The laser pulses within the scanning path have a significant impact on the ablation state of the micro-textures fabricated. As the scanning speed increases, the morphology of ablation craters changes from discrete craters to continuous overlapping craters due to the plume shielding effect. The plasma plume with a short formation time [[59], [60], [61]] generates substantial recoil pressure, expelling molten material away from the laser beam center, then ejected molten material subsequently redeposits onto the surface, forming a layer of resolidified debris [62]. When the plume shielding effect is neglected, the ablation depth of AlN by a nanosecond laser pulse increases with higher laser energy density. Conversely, when the plume shielding effect is considered, as the energy density gradually approaches the ablation threshold, the morphology transitions from discontinuous intermittent depressions to continuous overlapping ablation craters. This reduction in laser energy absorption on the ceramic surface is attributed to the irregular edges of craters. As the laser processes successive craters, energy accumulation during repetitive cycling enlarges adjacent craters, resulting in bottleneck-shaped ablation profiles along the scanning path. In addition, when the scanning speed surpasses the critical value, the shielding effect will be weakened, which is conducive to removing the ridge-like protrusions within the ablation trajectory and generating gradually uniform micro-grooves. The primary factors diminishing ablation efficiency with increasing laser energy density and pulse number are plasma formation and its shielding of the target. For ultraviolet radiation, the dominant mechanisms governing radiation-ablation interactions are multiphoton ionization and multiphoton absorption.

## 2.2. Different types of laser processing of ceramic surfaces

In laser manufacturing applications, beam quality and temporal-spatial characteristics critically determine processing outcomes. The wavelength of the laser beam covers various bands from ultraviolet to infrared, the wavelength of 180~400nm is the ultraviolet laser, the wavelength of 400~700nm is the visible laser, and the wavelength of 700~1000nm is the infrared laser. Materials have different absorption rates for laser beams with different wavelengths. Furthermore, based on the pumping mode, lasers can be classified into continuous lasers and pulse lasers. According to the pulse duration,

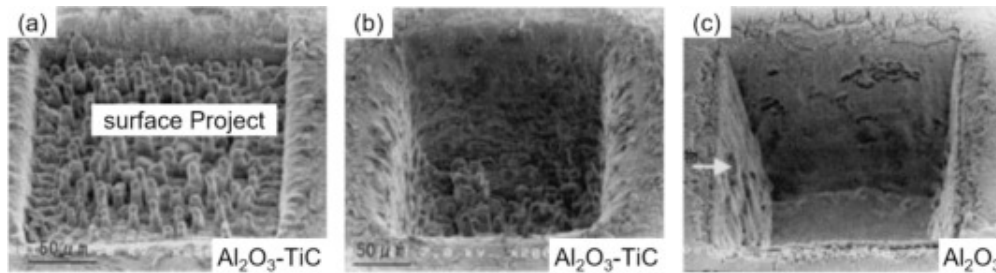
pulse lasers can be further categorized into quasi-continuous wave lasers (millisecond, microsecond), short pulse lasers (nanosecond) and ultra-short pulse lasers (picosecond, femtosecond). In general, the shorter pulse time, the higher single pulse energy, the narrower pulse width, and the higher processing accuracy. Specially, femtosecond laser processing operates through a non-thermal mechanism, wherein material transitions directly from solid to plasma state upon irradiation, thereby largely eliminating the formation of a recast layer. This unique characteristic enables its extensive application for precision microfabrication, including internal stealth processing of materials. Nanosecond lasers significantly reduce equipment costs and have been widely employed in selective metallization of ceramic microstructures.

Based on gain medium composition, lasers are categorized into four primary types: namely solids (crystals such as Nd:YAG, titanium-doped sapphire), liquids (such as organic dye solutions and inorganic compound solutions containing rare earth metal ions), gases (such as atoms, molecules, ions, excimer, etc.) and semiconductor lasers. Notably, excimer lasers, CO<sub>2</sub> lasers, and solid-state variants have demonstrated efficacy in ceramic surface modification, particularly for metallic layer formation as detailed in subsequent sections.

### 2.2.1. Excimer laser

Characterized by short wavelengths (193---351 nm) and correspondingly high photon energies (3.5---6.4 eV), excimer laser photons exhibit strong absorption in ceramic materials such as AlN and Al<sub>2</sub>O<sub>3</sub>. Affirmatively, specific excimer laser wavelengths demonstrate superior absorption characteristics in AlN ceramics compared to infrared laser wavelengths. Shafeev et al. [63] developed a two-stage laser-assisted metallization process without additional precursors, utilizing UV excimer laser ablation of ceramic surfaces and then depositing the metal via chemical methods. Their findings demonstrate that laser ablation modifies insulator band structures, generating a non-zero density state with energy approaching the metal reduction potential. Complementary research by Georgiev et al. [64] revealed that when the energy density of excimer laser exceeds

$1.0\text{J}/\text{cm}^2$ , the surface of AlN ceramics undergoes oxidation and decomposition, generating metallic Al and simultaneously forming a sub-surface layer tens of nanometers thick. These photochemical and photothermal transformations are accompanied by significant photomechanical damage in the thin surface region. IIDA et al. [65] employed KrF excimer laser irradiation for surface treatment of  $\text{Al}_2\text{O}_3$ -TiC substrates. The modified surface developed numerous conical structures with approximately  $10\mu\text{m}$  diameters and  $30\mu\text{m}$  heights, as illustrated in Fig. 6. However, the current excimer laser directly processes the ceramic surface, resulting in an insufficient thickness of the metal layer and an increase in resistivity. Furthermore, these laser systems exhibit inherent limitations including large footprint, high capital costs, and significant maintenance requirements [66], rendering them unsuitable for large-scale industrial metallization of AlN ceramics.



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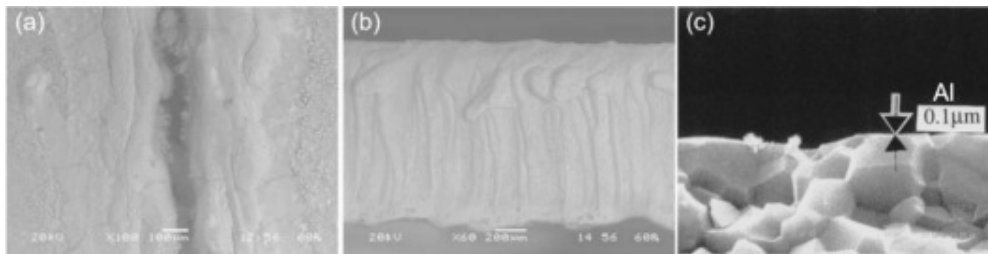
Fig. 6. Surface project of  $\text{Al}_2\text{O}_3$ -TiC and  $\text{Al}_2\text{O}_3$  ablated by KrF excimer laser: (a)  $\text{Al}_2\text{O}_3$ -TiC ( $14.8\text{ J}/\text{cm}^2$ , 200 pulses); (b)  $\text{Al}_2\text{O}_3$ -TiC ( $14.8\text{ J}/\text{cm}^2$ , 2000 pulses); (c)  $\text{Al}_2\text{O}_3$  ( $114.8\text{ J}/\text{cm}^2$ , 2000 pulses) [65].

### 2.2.2. $\text{CO}_2$ laser

$\text{CO}_2$  lasers with a wavelength of  $10.6\mu\text{m}$  exhibit high laser-to-optical conversion efficiency during material processing. The heat generated during laser operation can be efficiently dissipated from the gain region through convection or diffusion, enabling effective removal of surface materials, as illustrated in Fig. 7. Yabe et al. [67] fabricated



conductive traces on AlN surfaces using CO<sub>2</sub> laser lateral excitation, achieving the widths of 500 μm with electrical resistance below 0.1 Ω/mm. Owing to the advantages including rapid processing speed and low operational costs, CO<sub>2</sub> laser has been widely adopted as a diamond machining alternative in applications such as laser welding [68], surface modification [69] and medical treatment [70]. Molian et al. [71] implemented a scattered continuous-wave CO<sub>2</sub> laser technique for AlN ceramic processing, leveraging thermally induced stress from localized expansion/contraction gradients to achieve high-precision material removal with exceptional processing efficiency and quality. While CO<sub>2</sub> laser irradiation produces conductive layers with moderate thickness and acceptable resistivity for circuit patterning under optimal conditions, challenges persist regarding suboptimal edge definition in metallized regions.



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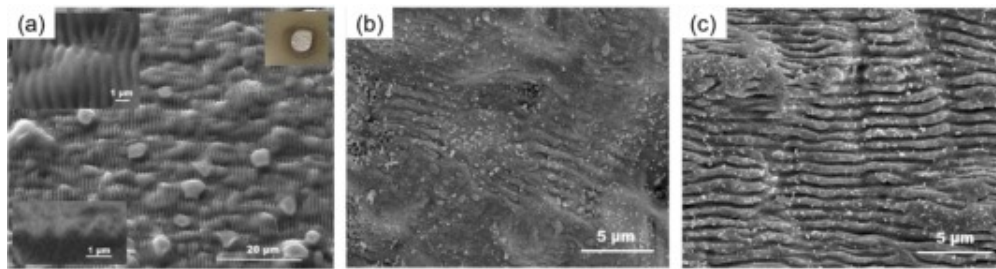
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Fig. 7. CO<sub>2</sub> laser processing of AlN: (a) Upper surface diagram, (b) Profile [71]; (c) SEM diagram of Al layer prepared [67].

### 2.2.3. Solid state laser

Solid-state lasers employ crystalline or glass host materials, which serve as the host medium and are typically doped with specific ions known as activating ions. Varying the dopant species within identical crystal matrices yields distinct lasing characteristics. The Nd: YAG laser utilizes an yttrium aluminum garnet (Y<sub>3</sub>Al<sub>5</sub>O<sub>12</sub>) host doped with Nd<sup>3+</sup> ions, capable of single-pulse operation while exhibiting superior pulse energy, beam quality, repetition rate stability, and continuous-wave performance. This system enables precise

and controllable material modifications [72], where in the deposited incident energy induces localized phase transitions and rapid thermal cycles, thereby altering the morphology and physicochemical properties of processed materials. Nedyalkov et al. [73] used Nd:YAG laser to obtain an aluminum thin layer with a period of 830nm and a conductive ripple structure on the ceramic surface, as depicted in the Fig. 8. Moreover, the Nd: YAG laser also can minimize uncontrolled material combustion during processing, enabling high-precision cutting and allowing for the machining of thicker materials at the same average power.



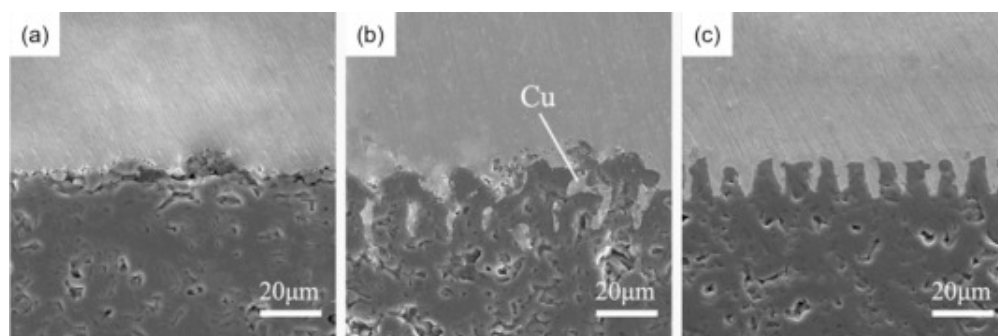
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Fig. 8. Machining of AlN ceramic surface by Nd: YAG laser [73]: (a) Ablation zone after machining by  $18\text{J}/\text{cm}^2$  and 900 laser pulses. Enlarged image of corrugated structure (top left), cross section view (bottom left), irradiation spot with diameter of about 2mm (top right); (b)  $20\text{J}/\text{cm}^2$ , SEM image after 100 laser pulses processing (c)  $32\text{J}/\text{cm}^2$ , SEM image after 100 laser pulses processing.

Mohammed et al. [74] investigated the influence of laser energy and pulse overlap on microchannel depth ( $200\mu\text{m}$ ) in ceramics, revealing an optimal peak fluence range of  $125\sim 130\text{J}/\text{cm}^2$ . Compared with Nd: YAG laser, Nd: YVO<sub>4</sub> laser has a wider absorption bandwidth and a larger stimulated emission cross section. Lv et al. [75] used a 355nm Nd: YVO<sub>4</sub> ultraviolet laser to directly modify the ceramic surface, as shown in the Fig. 9, and proved that the high aspect ratio microstructure formed by laser modification is an important reason for improved adhesion. The output wavelength of solid-state laser is

shorter than that of CO<sub>2</sub> laser, which is more conducive to the absorption by ceramic materials. Solid-state lasers are most suitable for industrial production and can form the thickest metal layers. However, the Gaussian beam characteristics of solid-state lasers result in relatively low the quality of metallized Al wire. How to prepare high-quality aluminum wires on the ceramic surface is still in the development stage and has become an important research topic.



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Fig. 9. Cross section of Nd: YVO<sub>4</sub> laser processing surface and copper plated ceramic [75]: (a) 3.8J/cm<sup>2</sup>, 250mm /s, 10µm; (b) 22.5J/cm<sup>2</sup>, 50mm/s, 15µm; (c) 11.3J/cm<sup>2</sup>, 50mm/s, 12.5µm.

In this paper, all three types of lasers have been applied in the surface microstructure processing and metallization of ceramic materials, and their advantages and disadvantages were summarized in the Table 2. To be specific, solid-state lasers have the advantages of low price, compact equipment structure, stable performance and high efficiency, and are mostly applied in the preparation and modification of the ceramic material microstructure to enhance the adhesion of subsequent coatings [76]. The fiber-delivery capability of Nd: YAG lasers can be integrated with robotic systems to create flexible processing platforms, thereby eliminating the need for complex beam transmission setups. This operational advantage provides significant competitiveness in ceramic surface processing applications. Fiber transmission can also compress the laser

pulse width through Q-switching to increase the peak power, making Nd: YAG lasers more suitable for preparing micro-nano microstructures on ceramic surfaces. In fact, laser ablation technology is increasingly being applied in the manufacturing of ceramic circuit boards, especially in electronic packaging. For instance, through laser processing, the precursor films of ceramics can be decomposed and their surfaces modified, thereby obtaining functional ceramic surfaces, which offers a promising approach to enhancing the bonding performance of ceramics.

Table2. A list of relevant research on various ceramic laser processing techniques.

| Type                       | Wavelength            | Pulse width | Material           | Optimization technology                    | Result                                                                                                              |
|----------------------------|-----------------------|-------------|--------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Excimer laser [63]         | 510nm                 | 10ns        | Insulator surfaces | Two-stage metallisation                    | The technology is flexible, but the graphics are difficult to be refined.                                           |
| Excimer laser [64]         | 248nm                 | 25ns        | AlN                | Plasma-source molecular beam epitaxy       | AlN can be decomposed, but it exhibits a limited thickness and is susceptible to considerable light-induced damage. |
| CO <sub>2</sub> laser [67] | CO <sub>2</sub> laser | CW          | AlN                | Control the background oxygen gas pressure | Higher speed and lower cost, but the metallization distribution exhibits unevenness.                                |
| CO <sub>2</sub> laser [71] | CO <sub>2</sub> laser | CW          | AlN                | Use oxygen as the assist gas               | Improve the accuracy and cutting speed, but the thermal stress is large.                                            |
| Solid-state laser          | 1064nm                | CW          | AlN                | In vacuum                                  | High-precision cutting is achieved, but the film surface is                                                         |

| Type              | Wavelength | Pulse width | Material                       | Optimization technology  | Result                                                                         |
|-------------------|------------|-------------|--------------------------------|--------------------------|--------------------------------------------------------------------------------|
| [73]              |            |             |                                |                          | prone to oxidation.                                                            |
| Solid-state laser | 1064nm     | 10 $\mu$ s  | Al <sub>2</sub> O <sub>3</sub> | Change the pulse overlap | The microchannel has high quality, but its geometry requires higher precision. |
| [74]              |            |             |                                |                          |                                                                                |
| Solid-state laser | 355nm      | 40ns        | Al <sub>2</sub> O <sub>3</sub> | Change the laser density | The adhesion capacity has been enhanced.                                       |
| [75]              |            |             |                                |                          |                                                                                |

### 3. Effect of laser processing on physical and chemical properties of ceramic surface

The deposition of suitable catalytic seeds on ceramic surfaces is a fundamental prerequisite for successful electroless plating. Palladium (Pd), owing to its unique catalytic activity, is highly valued and frequently used to initiate the electroless deposition reactions. However, traditional PD-based catalysts cannot achieve selective activation, and subsequent circuit imaging requires complex lithography processes [77]. Therefore, it is necessary to develop an effective method for selective preparation of high adhesion copper circuits. Selective laser ablation on ceramic substrate can achieve surface metallization and enhance the surface conductivity of the etched area. Laser ablation of ceramic surfaces is non-contact, environmentally friendly (no toxic chemical waste), and highly precise. These advantages facilitate precise control of circuit board dimensions and optimization of circuit architectures, making this technology an ideal choice for microfluidic circuit fabrication [1]. The metallization of ceramic surfaces can induce a series of changes in the physical and chemical properties, including surface morphology, chemical composition, and wettability. These alterations may be influenced by variations in laser ablation parameters, such as laser energy density, frequency, scan spacing, laser

energy distribution, and line spacing, thereby enabling the formation of microstructures on the ceramic surface that serve as catalytic active centers. Selective copper deposition is achieved within the laser-activated regions [78]. According to recent studies, surface morphology, chemical composition and wettability significantly affect the adhesion strength between electroless copper plating and copper layer, which will be discussed in detail in the following chapters.

### 3.1. Influence of laser processing parameters on ceramic surface microstructure

During the laser ablation process, the ceramic undergoes rapid melting, vaporization, and immediate rapid cooling of ceramics resulting in a variety of micro-scale surface morphologies on the ceramic surface, such as grooves, holes, cracks and recast layers. Laser activation decomposes the localized precursor film on the ceramic surface, with the decomposition products embedded in the substrate to catalyze electroless plating. Research demonstrates that laser surface modification significantly enhances coating adhesion [79]. The engineered surface topography not only increases interfacial contact area but also promotes favorable interfacial stress distribution, thereby improving the mechanical properties of treated materials.

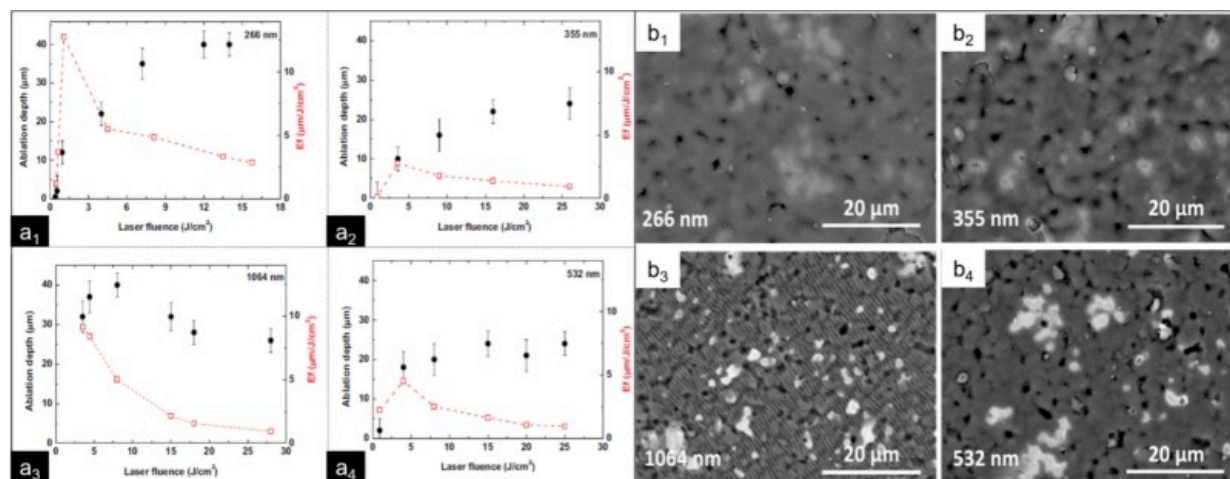
The micro and nano structures on the material surfaces fabricated by laser can significantly improve the mechanical, chemical and optical surface properties [77]. Different laser energy distributions have a significant impact on the surface morphology [80]. When uniform distribution is adopted, the solid–liquid boundary of molten pool is relatively flat. But when Xu [81] et al. adopted Gaussian laser energy distribution for processing, there was a temperature gradient on the material surface, and the molten pool will flow from the center to the edge, resulting in obvious surface deformation, that is, the Gaussian distribution laser was more conducive to the removal of the material as the heating time increased. Besides, laser ablation can be used to produce complex patterns on ceramic surfaces, thus providing some novel and special functional surfaces. N. edyalkov et al. [82] used nano-laser to process AlN surface and studied the influence of laser energy density and pulse number on ablation depth and morphology of ablation



zone. The ablation depth is a function of the laser intensity and pulse number, and the ablation efficiency decreases with the increase of the laser intensity and the pulse number.

The various micro-nano structures processed on the material surface by laser treatment also contribute to the formation of the surface conductive layer. Fig. 10 (a<sub>1</sub>~a<sub>4</sub>) shows the relationship between ablation depth and laser intensity. At the wavelengths of 532nm, 355nm and 266nm, the ablation depth per flux  $E_f$  ( $\mu\text{m}/\text{J}/\text{cm}^2$ ) firstly increases and then decreases with the increase of irradiation amount. It is noteworthy that at a wavelength of 1064nm, the ablation zone near the ablation threshold exhibits a wavy surface structure, and the thickness of formed layer increases from several nanometers to hundreds of nanometers. As the laser fluence approaches the ablation threshold of  $28\text{J}/\text{cm}^2$  with 100 pulses applied, a more uniform corrugated structure is generated, featuring an average size of approximately  $20\mu\text{m}$ , as illustrated in Fig. 10 (b<sub>1</sub>–b<sub>4</sub>). This structured surface demonstrates favorable electrical conductivity. Radhakrishnan et al. [83] successfully fabricated high-performance Ox-Ox ( $\text{Al}_2\text{O}_3/\text{Al}_2\text{O}_3$ ) ceramic composites using picosecond laser processing. During ultrafast laser interactions, energy transfer predominantly occurs via electron dynamics, while laser energy absorption is governed by the bandgap energy of the material. At higher scanning speeds, material removal rates (MRR) are primarily ablation-dominated, whereas lower scanning speeds promote melt-dominated removal mechanisms. Excessive heat input induces molten pool formation with concomitant surface oxidation. Consequently, optimal MRR is achievable only when critical equilibrium exists between ablative and melt-driven material removal pathways. Recently, N. Edyalkov et al. [84] investigated laser ablation of AlN ceramic under six distinct ambient pressures ranging from 1 to  $10^{-4}$  atm, and Fig. 11 demonstrates that ripple structures emerge at a fluence of  $10\text{J}/\text{cm}^2$  with 20 pulses. When the number of pulses increases to 600, periodic structures with an average period of approximately  $20\mu\text{m}$  are obtained, and the ridge-like nanostructures transform into oriented periodic microstructures (OPMS). Fig. 12 further reveals that the formation of OPMS requires exceeding a certain threshold. The surface of OPMS can conduct electricity due to the

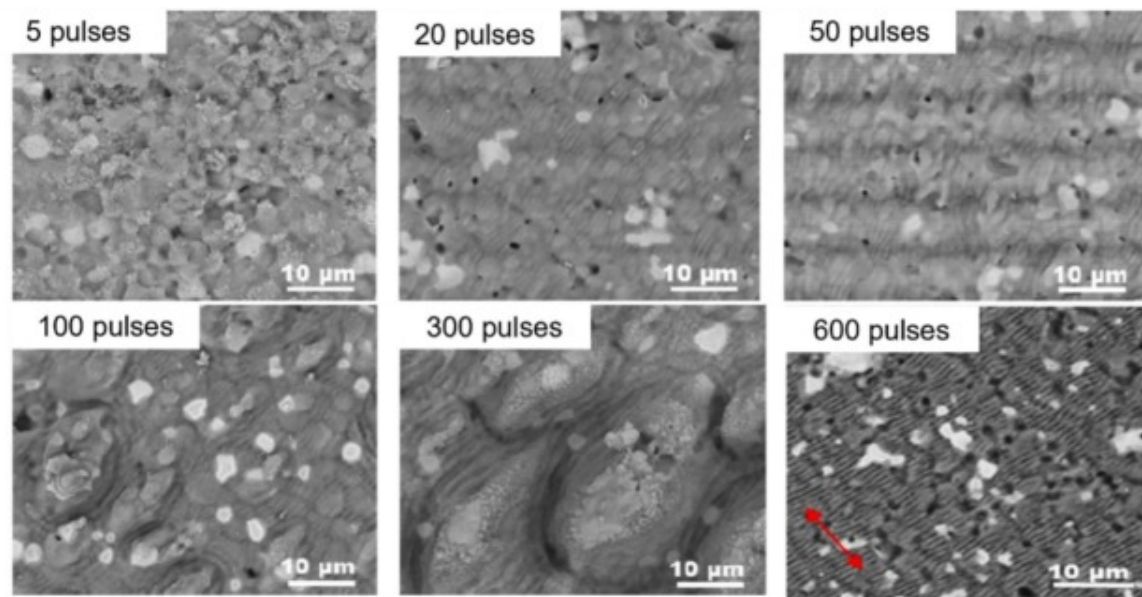
activated metallic aluminum, with electrical conductivity ranging from several to tens of ohms. At a fluence of  $6.7 \text{ J/cm}^2$ , microstructural orientation becomes discernible, but within the range of  $8\text{--}13 \text{ J/cm}^2$ , the peak height of the periodic structure decreases significantly and the periodic structure will disappear at  $15.5 \text{ J/cm}^2$ . In other words, periodic structures can be regulated by laser energy density. Defect induction, energy transfer, and shock-induced sputtering during nanosecond laser processing significantly influence laser ablation mechanisms. In addition, defect induction, energy transfer, and shock-induced sputtering during nanosecond laser irradiation play critical roles in the laser ablation process. To investigate these mechanisms, Li et al. [85] numerically simulated the filamentation behavior and ablation morphology of alumina ceramics irradiated by femtosecond lasers under two different pulse energies. The results show that the filament length ablated with a 4 mJ pulse energy and a 500 mm lens can be extended up to 25 mm, forming an ablation hole with a diameter of  $340 \mu\text{m}$  and a depth of  $440 \mu\text{m}$  along the middle part of the filament. The power clamping effect brought by the constant filament diameter and plasma intensity ensures uniform morphology and size of the ablation holes across the entire filament, and the plasma shock wave can effectively remove the recast layer on the hole wall.



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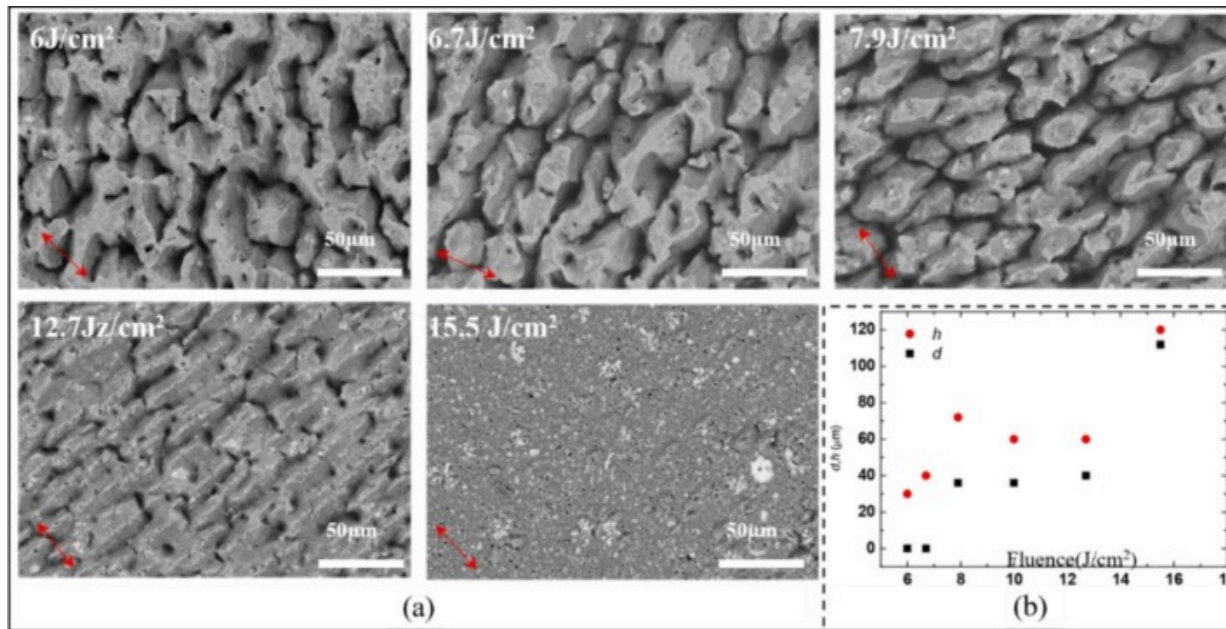
Fig. 10. Ablative depth as a function of the laser radiation at four wavelengths used: ( $a_1$ ) 532nm; ( $a_2$ ) 355nm; ( $a_3$ ) 266nm; ( $a_4$ ) 1064nm; SEM images of the ablation area surface at four wavelengths used: ( $b_1$ ) 266nm; ( $b_2$ ) 355nm; ( $b_3$ ) 1054nm; ( $b_4$ ) 532nm [80].



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Fig. 11. Effect of different pulse numbers on the surface morphology of vacuum laser ablation of AlN ceramics [81].



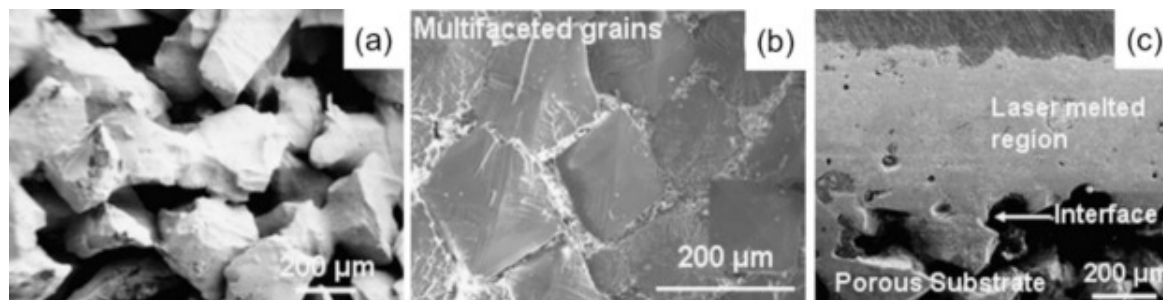
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Fig. 12. Effect of laser energy density on the surface morphology of vacuum laser ablation of AlN ceramics: (a) SEM images of the surface morphology obtained under different effects of 600 pulses; (b) Dependence of parameters  $d$  (depth at the top of the wave crest) and  $h$  (valley depth relative to the initial surface) on the laser energy density [81].

In addition to SEM characterization of grain morphology and crystalline structures in laser-ablated ceramics, surface roughness and porosity provide quantitative metrics for statistical analysis of topographical evolution. Harimakar et al. [86] found that as the laser dosage increased, the surface roughness significantly decreased from 3.49 μm to 1.07 μm by quantifying surface roughness. This reduction is attributed to angular grain growth and progressive grain flattening under higher energy densities, ultimately forming high-density surface structures as evidenced in Fig. 13. Furthermore, material removal rates correlate strongly with laser beam intensity, where higher intensities yield greater ablation efficiency. Umer [87] et al. used Nd: YAG laser with wavelength of 1064 nm to

carry out laser micro-machining on the  $\text{Al}_2\text{O}_3$  surface. The laser intensity from low to high ( $L_1=85\%$ ,  $L_2=90\%$ ,  $L_3=95\%$ ) was selected, along with three pulse frequencies ( $P_1=4$  kHz,  $P_2=6$  kHz,  $P_3=8$  kHz), pulse durations from short to long ( $T_1=2\mu\text{s}$ ,  $T_2=4\mu\text{s}$ ,  $T_3=6\mu\text{s}$ ), and pulse overlap rates varying from small to large pulse ( $PO_1=33.33\%$ ,  $PO_2=50\%$ ,  $PO_3=66.66\%$ ). Fig. 14a is the surface topography of  $\text{Al}_2\text{O}_3$  before processing, while Fig. 14b and 14c represent the best and worst surface morphologies of laser-processed ceramic surfaces, respectively. Especially in Fig. 14c, the surface morphology is relatively rough, and there are many gaps between the grains. At this time, the pulse over-lap degree reaches 66.66%, which means that high pulse overlap indicates high energy per unit area. This also results in a deeper depth of each scan and a very rough surface. Fig. 14d shows the relationship between the surface roughness of  $\text{Al}_2\text{O}_3$  ceramics and the variables. It can be observed that both pulse intensity and overlap rate have nearly equal effects on surface roughness, each approximately twice that of frequency, while the impact of pulse duration is negligible.

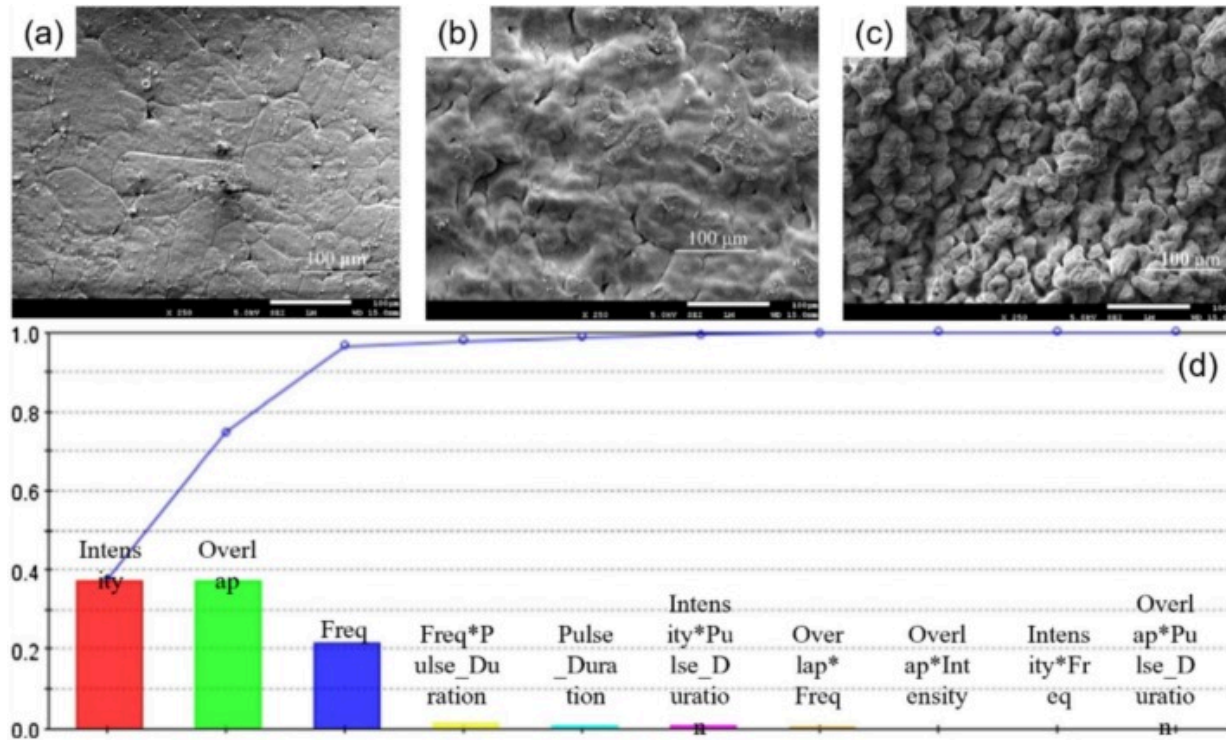


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Fig. 13.  $\text{Al}_2\text{O}_3$  surface microstructure: (a)  $\text{Al}_2\text{O}_3$  surface microstructure; (b) Surface microstructure after laser processing ( $573\text{J}/\text{cm}^2$ ); (c) cross-section diagram after laser processing ( $573\text{J}/\text{cm}^2$ ) [82].





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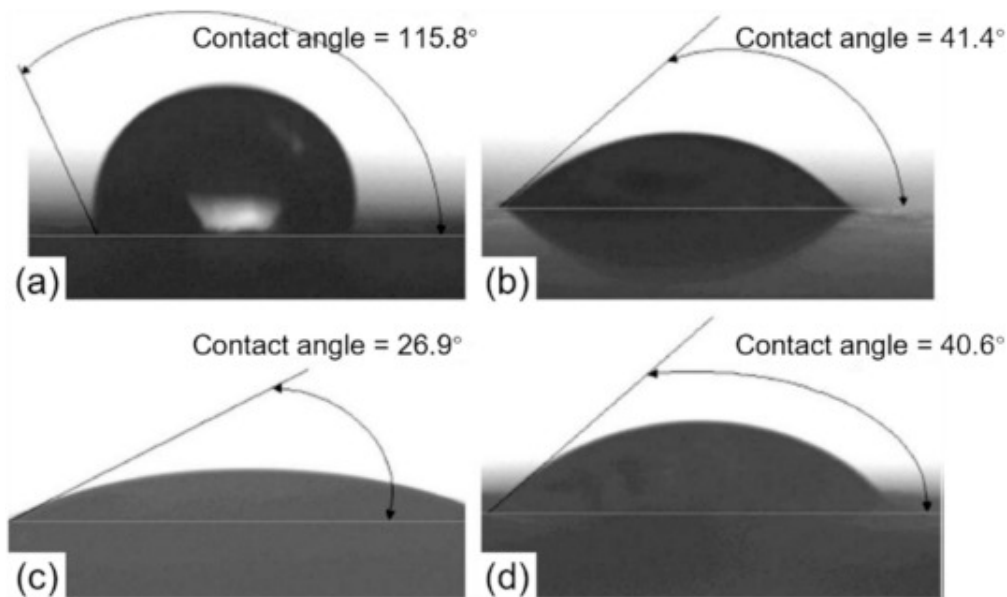
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Fig. 14.  $\text{Al}_2\text{O}_3$  sample: (a) Original sample before processing; (b) Optimal SEM image of the sample surface; (c) SEM images of the worst sample surface; (d) Relationship between surface roughness and input variables [87].

Notably, laser processing significantly alters the wettability of ceramic surfaces. The induced superhydrophilicity critically enhances interfacial bonding between substrates and functional materials, enabling applications such as cell-based biosensors, advanced coating fabrication, and water-assisted flow generation, as illustrated in Fig. 15 [88,89]. Cao et al. [90] developed circuit patterns on PEEK (Polyether Ether Ketone) substrates via electroless plating and ultrasonic cleaning by selectively altering the surface wettability. The deposited Cu layer featured a microcolumn array structure, which substantially



enhanced the contact area with the substrate. Kunz [91] et al. prepared periodic structures on the surface of  $\text{Al}_2\text{O}_3$ -n $\text{ZrO}_2$ -Nb (78.3–1.7–20vol%) ceramic matrix composites by laser, and the contact angle of the composite surface was reduced from  $68.4^\circ$  to  $40.9^\circ$ . Triantafyllidis et al. [92] studied the effect of laser surface treatment on the fluid contact angle of  $\text{Al}_2\text{O}_3$  based refractory ceramics. Compared with the treated surface, the untreated surface showed considerable heterogeneity in wetting and enhanced wetting (contact angle decreased from  $41.4^\circ$  to  $26.9^\circ$ ). Experiments show that laser power, surface roughness and surface wettability have a mutually reinforcing effect.



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Fig. 15. Laser machining ceramic surface wettability [88].

In laser ablation machining, the light and electrons interact through the principle of Fresnel absorption. If the laser intensity and interaction time are sufficient (the pulse width is greater than the lattice vibration response time), the lattice vibration intensifies, and the thin layer on the surface begins to melt, and eventually forming a metal melt

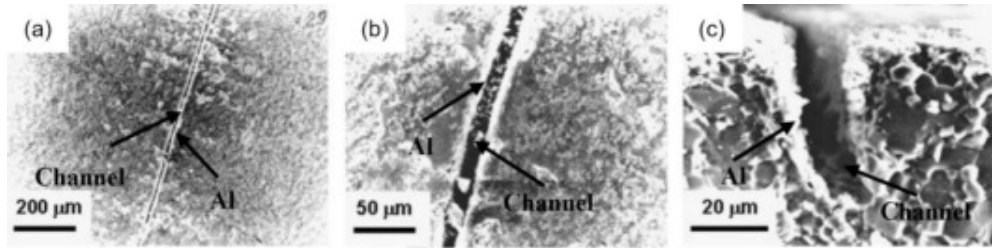
layer. The surface morphology of ceramics processed by laser is mainly related to laser beam parameters such as laser energy density, energy distribution, and pulse duration. As the energy accumulates, the temperature of the laser irradiation region exceeds the boiling point of the material, which will cause the surface of the material to melt or even evaporate. In the later stage of laser action, the melt on the laser surface begins to solidify on both sides of the scanning path, resulting in a more complex surface topography, and surface roughness is usually used for quantitative evaluation of surface topography. The changes in surface morphology and physical texture significantly affect various performance characteristics, so in addition to investigating laser processing parameters, factors such as scanning speed, scanning interval and pulse frequency also play a crucial role in the design of the microscale structure of laser-ablated ceramic surfaces, and selective removal of ceramic materials can be achieved by precisely controlling laser processing.

### 3.2. Effect of laser processing environment on ceramic surface microstructure

Surface engineering of materials can mitigate friction and regulate wear, thereby extending the service life of mechanical systems while enhancing their operational performance. In laser micro-machining, thermal accumulation around the laser-affected zone usually causes defects such as ripples and microcracks. Conversely, controlled heat management plays a crucial role in designing surface structures for laser-ablated ceramics. The occurrence of microcracks can be mitigated by optimizing laser parameters and controlling the processing environment. The characterization of microstructure and morphology is often carried out by using instruments or methods such as scanning electron microscopy (SEM), optical microscopy, X-ray diffraction (XRD), and nanoindentation testing.

SEM and XRD analyses ultimately demonstrated that high-energy laser irradiation disrupted the bonding of the surface AlN. When the temperature of the laser processing site rises to 2000-2700°C, the surface of aluminum nitride will decompose to release nitrogen gas and form conductive metallic aluminum, as shown in Fig. 16. But meanwhile,

a destructive factor in laser-induced selective metallization process is the presence of oxygen, which leads to the oxidation of Al on the irradiated surface, thereby increasing the resistivity of the structure [56].



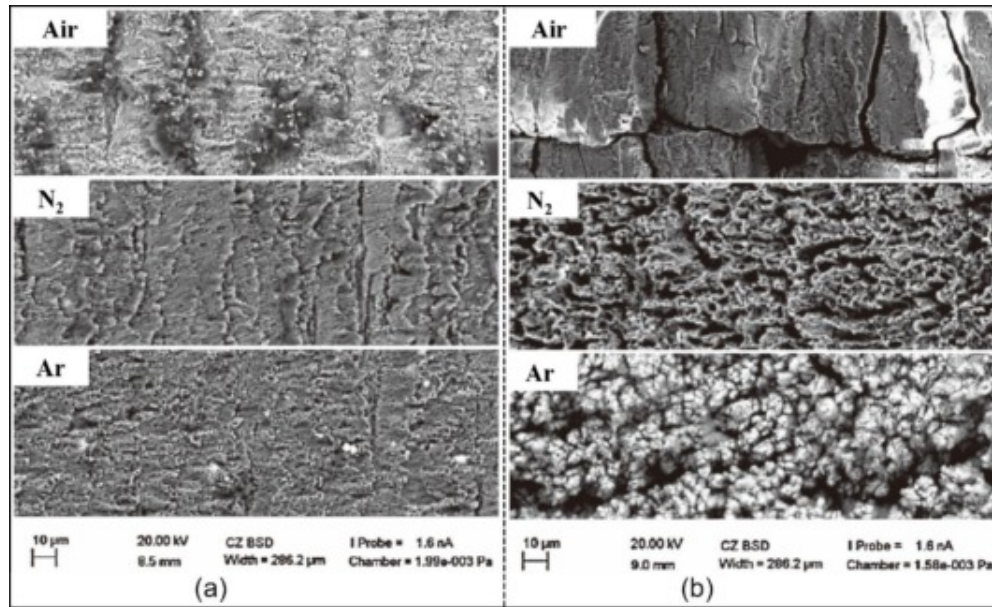
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Fig. 16. SEM image of laser processing AlN surface channel [56].

As illustrated in Fig. 16, due to the increase of sediment with the rise of pulse energy, the shape of the channel is affected by the thermal conductivity of material, and the upper width of the laser-processed AlN channel decreases. Koziol et al. [93] reported that laser treatment of AlN ceramic surfaces can produce metallic layers with thicknesses ranging from 3 to 10 μm under ambient air conditions, however, these layers often contain porous metal agglomerates because the precipitated Al readily reacts with oxygen to form Al<sub>2</sub>O<sub>3</sub>.

With the increase of energy, micro-cracks will occur in the metallized layer, while the size of the agglomerates generated under the protection of nitrogen and argon becomes larger and the porosity is more prominent. Fig. 17a shows the structure obtained by the laser flux close to the ablation threshold, and Fig. 17b shows the structure obtained by the laser flux far beyond the ablation threshold. Micro cracks appear on the machined surface in the air environment, while in the argon and hydrogen environment, the aggregate has a porous structure, and the Al<sub>2</sub>O<sub>3</sub> content is significantly reduced. In the inert gas argon environment, the resistivity  $\rho = 0.64 \times 10^{-6} \Omega$ , the thickness of the Al layer is 10 μm, and the laser flux is 640 J/cm<sup>2</sup>.



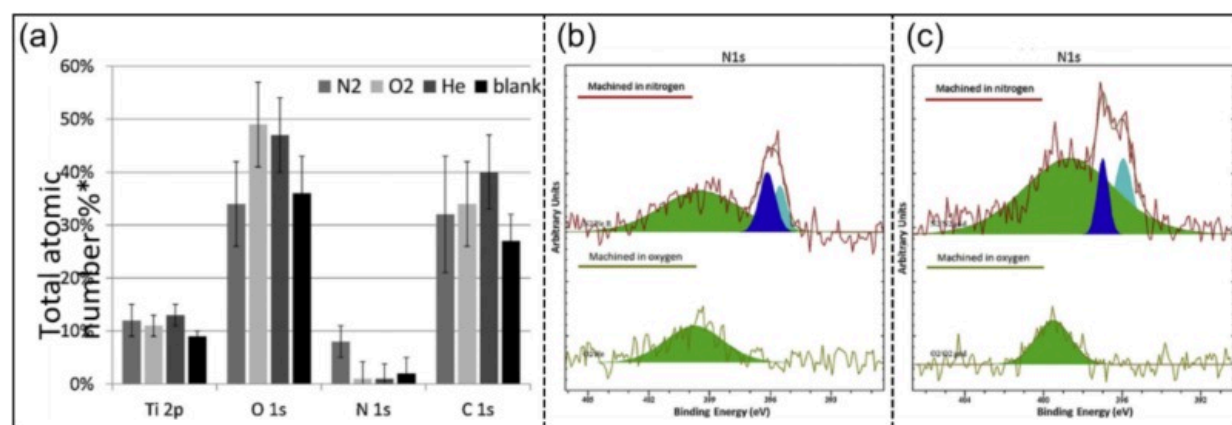
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Fig. 17. SEM images of AlN ceramic surface morphology under different atmospheres (Zeiss EVO MA 25): (a) The laser flux is lower than the ablation threshold; (b) Laser flux above the ablation threshold [92].

Lehr et al. [94] investigated the influence of different gaseous environments on the surface morphology of titanium processed by femtosecond laser. Fig. 18(a)-(c) present the elemental composition of sample surfaces under various processing conditions and corresponding XPS spectra before and after ablation. Concave-convex microstructures were generated by controlling combinations of laser fluence and pulse number. Samples processed in nitrogen exhibited higher N 1s peak intensity than those treated in other atmospheres, indicating enhanced formation of TiN which promotes osseointegration of metallic implants. Consequently, helium-processed samples demonstrated superior bonding strength and enhanced biocompatibility. Antończak et al. [54] formed a conductive structure directly on the AlN surface by laser induced decomposition, and

analyzed the influence of protective gases (air, argon, nitrogen) on the low ohmic structure by qualitative EDX analysis, as shown in Fig. 19.

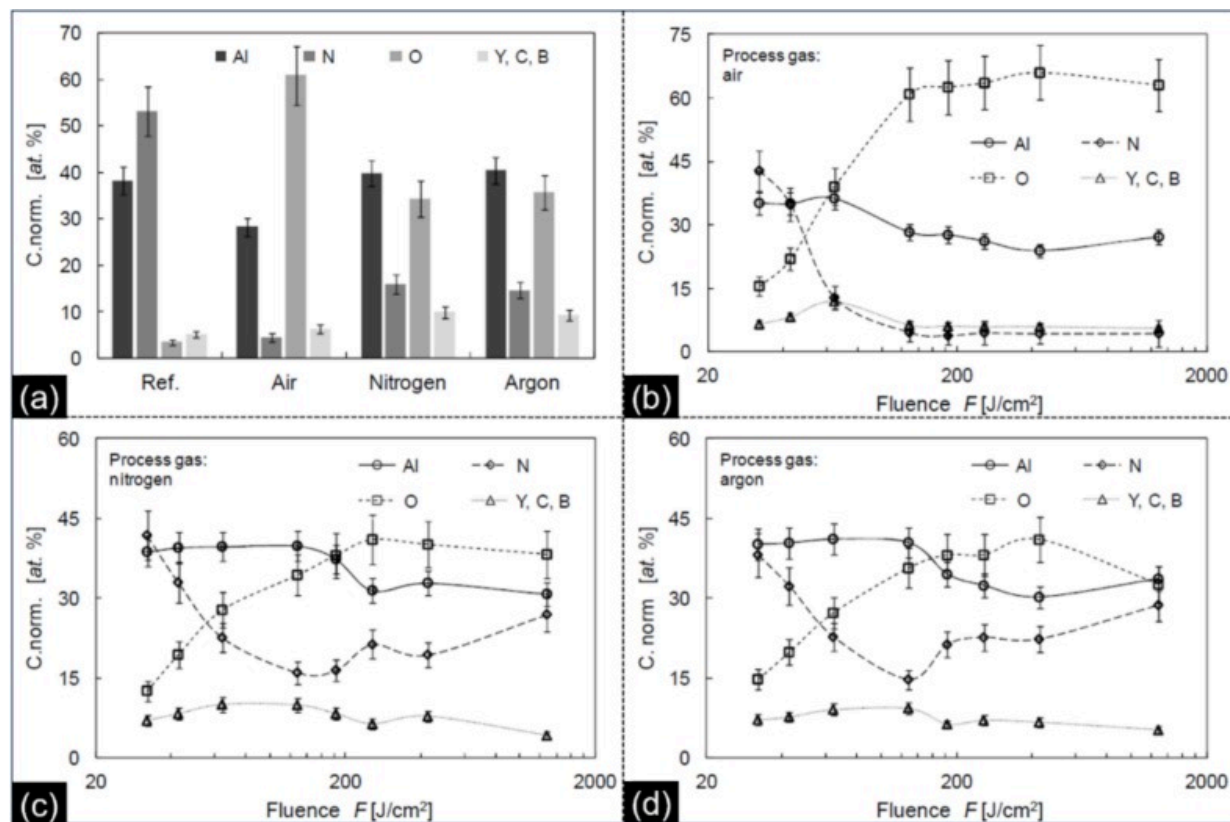


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Fig. 18. Effect of gas environment on surface chemical composition (a) Surface substance content determination of samples processed in nitrogen, oxygen and helium with XPS; (b) XPS spectra of N<sub>2</sub>-He and O<sub>2</sub>-He samples; (c) XPS spectra of N<sub>2</sub>-He and O<sub>2</sub>-He samples after ablation [94].





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Fig. 19. Chemical composition of the metallized layer: (a) Comparison of the results of various protective gases under the ablation threshold ( $128\text{J/cm}^2$ ); (b) – (d) Influence of the composition change and accumulation of various protective gases [54].

The results show that the Al content of the metallized layer increases significantly in the inert gas environment at the initial stage. With the increase of laser intensity, the nitrogen content of AlN on the surface decreases sharply in nitrogen from 53.1 at.% to 16 at.% and that in argon decreases to 14.7 at.% due to the reason of nitrogen shielding. When the pulse overlap rate reaches 94%, the path resistivity is  $0.64 \times 10^{-6} \Omega$ , which is only 20 times higher than that of pure Al ( $2.8 \times 10^{-8} \Omega$ ).



To sum up, the output beam of the solid-state laser is a circular Gaussian beam with high energy in the middle and low energy at the edge, resulting in poor metal density, low adhesion and small thickness at the edge. After increasing the energy, excessive intermediate energy will cause the metallized layer to undergo fission, increase the resistance, and decrease the adhesion. In particular, Al is highly reactive with oxygen at high temperatures, it is necessary to metallize in an inert gas environment. In addition, auxiliary gas is conducive to increasing the cutting depth. This is because the increase of auxiliary gas pressure accelerates the splashing rate of the sample melt and reduces the solidification of the melt on the cutting surface [2]. It should be noted that the pressure also cannot be increased indefinitely, which will bring dark cracks to the cutting process of ceramics. Moreover, the laser processing environment also has an important impact on the surface chemical modification of ceramic materials. In ambient air, higher laser power is required to achieve the necessary thermodynamic effect and remove the surface of the material [93]. In contrast, laser ablation in the inert gas environment enables direct removal of the ceramic surface at lower power levels while simultaneously forming a newly activated layer. This process thereby provides superior surface chemical conditions for subsequent bonding.

### 3.3. Effect of combined pulse sequence laser processing on ceramic surface microstructure

Ceramics have been widely used as electronic substrates due to their superior mechanical and physical properties. With the increasing demands of modern electronic technology, conventional mechanical methods or single pulse processing methods can no longer meet the industrial efficiency requirements for processing high-hardness ceramics [95,96]. On this basis, some researchers used single-mode millisecond laser combined with high-pressure gas jet to performed the rapid laser processing on the  $\text{Al}_2\text{O}_3$  ceramics [97,98]. On the one hand, this technology will cause the loss of laser energy and a large amount of gas consumption. On the other hand, the auxiliary gas pressure will change the refractive index of the medium between the workpiece surface and the nozzle, resulting in the defocusing of the laser spot. Therefore, in order to improve the efficiency and quality of

laser processing, Jia et al. [96] proposed a new combined laser pulse (CLPs) processing method consisting of millisecond pulses and auxiliary nanosecond pulses. Under the same total energy, the pulse peak intensity of the compound laser pulse trains is two orders of magnitude lower than that of an ordinary millisecond laser, it has been observed that the enhanced interaction between millisecond and nanosecond pulses facilitates faster formation of keyholes. Moreover, the recoil pressure generated by high-intensity laser pulse or the high shock pressure produced by plasma ignition can enhance the surface removal efficiency. Table 3 analyzes the comparison of processing quality between the composite laser pulse method, nanosecond-millisecond (ns-ms) two-step method and the conventional millisecond pulse method. It can be found that the diameter of micro-holes processed by conventional millisecond laser is approximately 160  $\mu\text{m}$ , which is much larger than the diameter of the laser spot. This is caused by the absorption coefficient variation during the processing. On the contrary, when the ns-ms two-component method and the composite laser pulse processing are adopted, the absorption efficiency of  $\text{Al}_2\text{O}_3$  remains at 0.8–1.0, which can effectively avoid the problem of absorption rate change and reduce the processing loss, and thus the aperture is closer to the spot diameter. The melt produced by the millisecond pulse can be ejected at velocities of tens of meters per second within a few milliseconds [100]. This rapid ejection helps to reduce the splatter deposited on the sample surface, thereby decreasing both the diameter on the upper surface and the taper angle of the holes. Further, to address the challenges of power density attenuation over long distances and low damage efficiency, Guo et al. [101] proposed a novel combined pulsed laser (CPL) technique based on the synergistic interaction between a femtosecond laser-induced air filament and a continuous-wave (CW) laser. By adjusting the position of the focusing lens, the focus of the CW laser can be respectively aligned with the front, middle, and rear segments of the femtosecond laser-induced filament, and the 3 mm thick alumina ceramic can be penetrated within 20 ms. It is indicated that this method not only enhances the absorption of laser by the target material, but also promotes efficient ejection of molten material via shockwave-induced sputtering. The femtosecond-CW CPL method not only

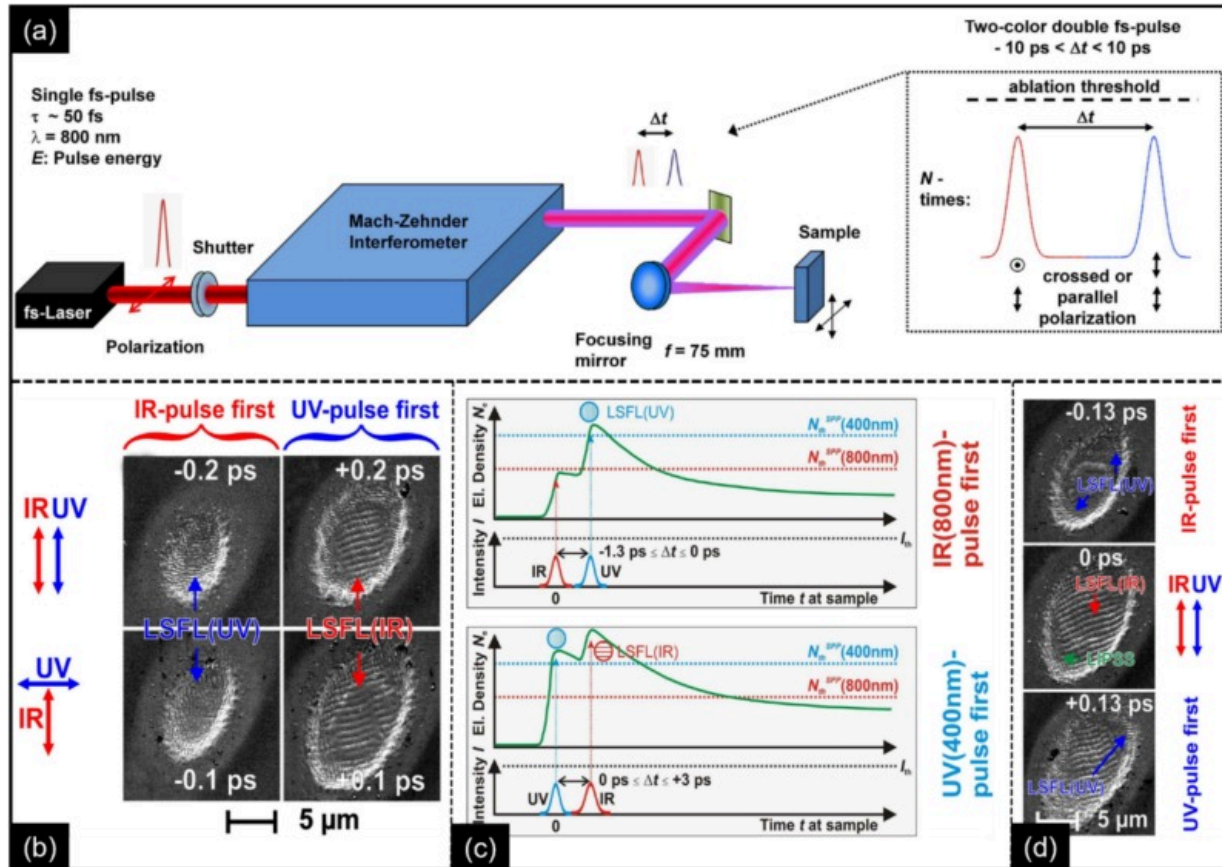
enhances the absorption of the laser energy by the target material, but also promotes efficient ejection of molten material via shockwave-induced sputtering.

Table 3. Drilling quality comparison between the CLPs, ns-ms two step and ms pulse method [99].

|                 | <b>Hole diameter (upper surface / lower surface)</b> | <b>Circularity (upper surface / lower surface)</b> | <b>Taper angle</b> |
|-----------------|------------------------------------------------------|----------------------------------------------------|--------------------|
| <b>CLPs</b>     | 95 $\mu$ m / 50 $\mu$ m                              | 0.96 / 0.95                                        | 2.6°               |
| <b>Two-step</b> | 120 $\mu$ m / 50 $\mu$ m                             | 0.96 / 0.92                                        | 4.0°               |
| <b>ms</b>       | 160 $\mu$ m / 90 $\mu$ m                             | 0.94 / 0.92                                        | 4.0°               |

The irradiation of solid materials with linearly polarized laser pulses commonly produces periodic surface structures or ripples. Due to the generated electromagnetic wave interference and excited plasma polaritons by the incident laser with the rough surface, the low spatial frequency periodic surface structures are often observed on strongly absorbing materials [[102], [103], [104]]. As shown in Fig. 20(a), Hohm et al. [105] used a Mach-Zehnder interferometer to generate parallel or cross-polarized double-sided impulse sequences at 400 and 800nm wavelengths. Fig. 20 (c) shows that the ultraviolet pulse that follows the red one can promote additional electrons into the conduction band. For a larger negative delay ( $\Delta t < -1.3$ ), the carrier density cannot be driven above the damage threshold, so the processed pattern is shown on the left side of Fig. 20 (b). Ultraviolet pulses will increase the carrier density, and the second infrared pulse reaches the surface and interacts with the active region of the previous carrier body. The periodic structure induced by laser occupies the central part of the surface morphology, as shown on the right of Fig. 20 (b). In other words, the coverage area of the LIPSS strongly depends on the sequence in which the ultraviolet/infrared pulses action. In addition, infrared and ultraviolet light sources are always perpendicular to the polarization direction and show

periods close to the corresponding wavelength. Consequently, current understanding posits that plasma-mediated mechanisms govern the formation of periodic surface structures on ceramics. As evidenced in Fig. 20(d), when time is 0, vertically aligned LIPSS emerge at the ablation pit periphery, exhibiting orientation parallel to the polarization vectors of both infrared and ultraviolet pulses. This alignment indicates interference phenomena arising from harmonic generation and frequency mixing during temporal overlap of time-overlapping pulses.



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Fig. 20. Surface structure of zinc oxide induced by two-color femtosecond laser (a) Experimental device based on Mach-Zehnder interferometer [106]; (b) Scanning electron microscope images of ceramic surface irradiated by NDPS=20 two-color two-FS pulse sequence with small positive and small negative delay; (c) SPP-based LIPSS formation scheme under dual-color dual-pulse ceramic; (b) Scanning electron microscopy of ceramic surface irradiated with NDPS=20 two-color dual-FS pulse sequence under three different delays [105].

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In summary, laser changes the physical and chemical properties of ceramic material surfaces, mainly reflected in the surface topography, chemical composition and wettability of three aspects. In addition, laser processing can also lead to the formation of various micro-nano structures on the surface of porcelain. Laser processing can also induce the formation of various micro-nano structures on the ceramic surface. However, high laser flux can induce the formation of oxygen vacancies, resulting in the activity of chemical gilding and the rough surface morphology of deposited particles.

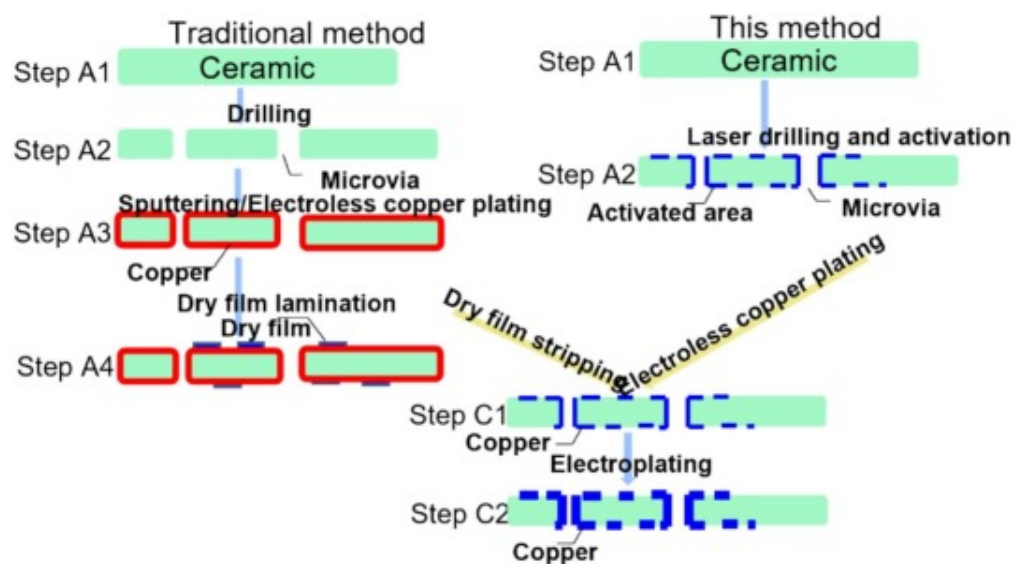
## 4. Electroless copper plating on AlN ceramic surface

### 4.1. Principle of electroless copper plating

Traditional ceramic metallization primarily relies on deposition techniques in vacuum involving metal particle evaporation [107], sputtering [108], and chemical vapor deposition [109] to form uniform metallic coatings on surfaces. However, these methods require complex multi-stage photolithographic procedures, prolong the metallization and increase implementation costs. Electroless copper plating still maintains play a crucial role in metallization processes due to its cost-effectiveness, low-temperature operation, and high throughput [110]. This technique can empower non-conductive substrates, such as glass, polymers, ceramics, and resins to enable electrical and electronic applications. Nevertheless, the dense and smooth surface of ceramics can hinder the effective adhesion of copper, so the pretreatment process before electroless copper plating is indispensable [[111], [112], [113]]. The pretreatment procedures generally involve roughening the



ceramic surface in an acidic or alkaline solution using laser irradiation, sandblasting, or chemical etching, followed by surface activation using catalysts such as Pd, Au, or Ag [114,115]. Although many studies have shown that the adhesion of copper layers is constantly improving, the current pretreatment methods for ceramics are still complex and inefficient, which seriously limits the practical application of metallization. Recently, Yung et al. [116] modified the surface of AlN ceramics with nanosecond lasers and then controlled the thickness of the copper film through chemical deposition without external current. It can be found from Fig. 21 that the thickness of the copper coating is closely related to the laser energy, reaching its maximum value when the laser energy approaches the ablation threshold. This part primarily focuses on the effects of laser and plating solution parameters on the film fabrication process, as these factors significantly influence the physical, chemical, and electrical properties of the coating.



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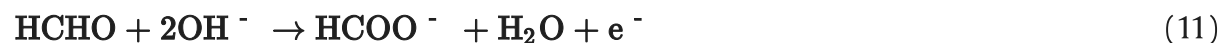
Fig. 21. Comparison between traditional method and laser activated metallization method [116].



Laser-induced activation and subsequent metallization in a chemical copper bath of fine lines and micropores on an AlN substrate is a favorable additive and precursor free method. Using formaldehyde (HCHO) as reducing agent, the overall reaction of ECP is as follows:



Among them, HCOO<sup>-</sup> (formic acid) is a reducing agent of the acid oxidation products. Equation (9) is an electrochemical model. Wagner and Traud used the mixed potential theory to explain metal corrosion process, and Paunovic and Saito applied this theory to the explanation of copper chemical deposition. Meanwhile, the total reaction given in Equation (9) can be decomposed into a simple reduction reaction, a cathode reaction, an oxidation reaction, and an anode reaction.



Therefore, the entire reaction (9) results from the combination of two distinct half-reactions (7), (8), both of which occur at the same electrode within the metal-solution interphase. The characteristic of these semi-reactions their oxidation and reduction electrode potentials. However, the redox potential is related to the Gibbs free energy change ( $\Delta G$ ) and the electrode reaction rate constant (K) parameters, which can predict the trend of the ECP reaction [117]. This approach not only streamlines the ECP process and reduces costs, but also yields high-performance conformal coatings, establishing it as a highly promising maskless microfabrication technique.

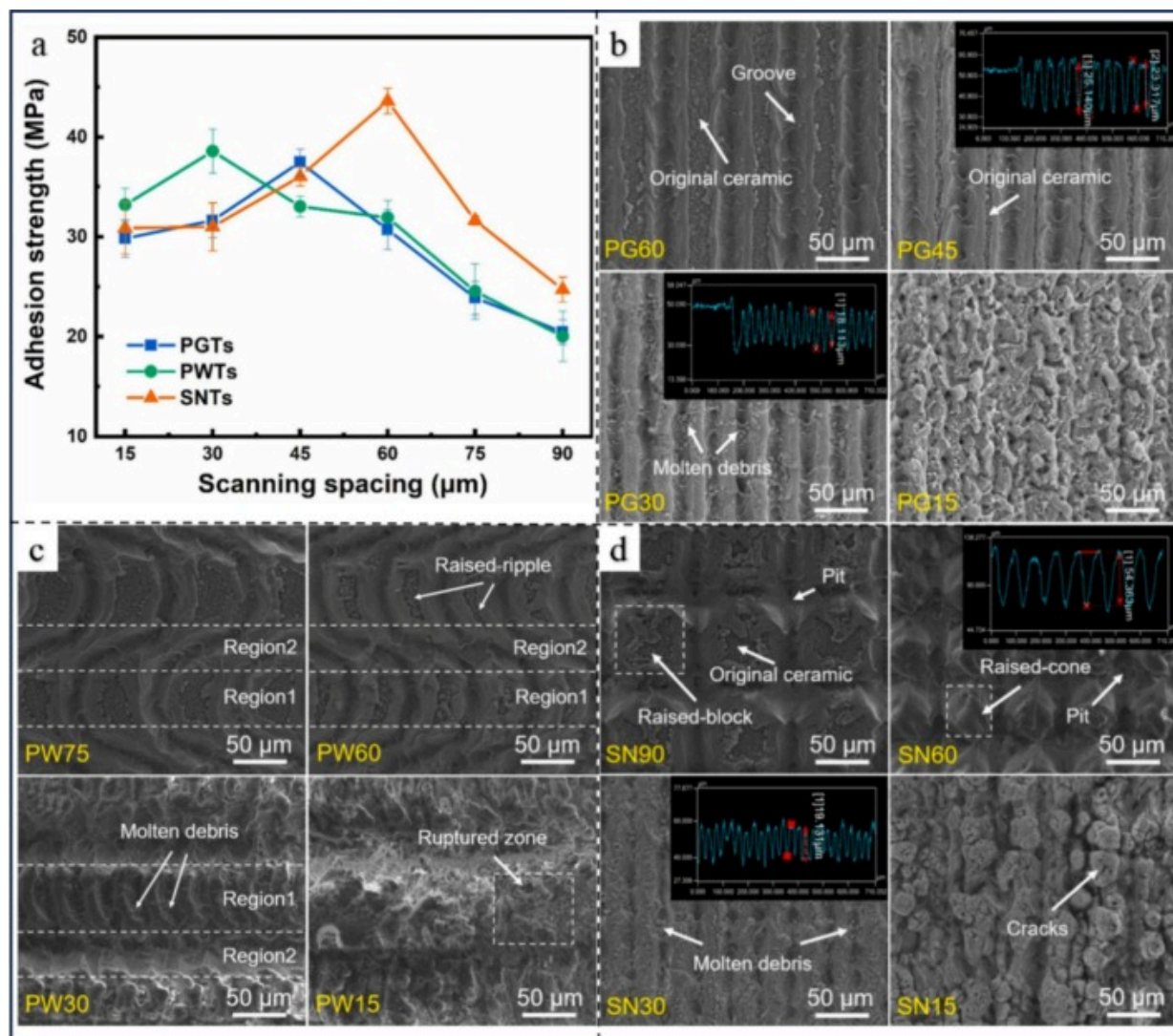
The reduction reaction of copper ions ( $\text{Cu}^{2+}$ ) plays a crucial role in electrochemical deposition, metallization of electronic devices, and energy conversion systems, due to its ability to precisely control surface structures and electronic properties. In recent years, this capability has been leveraged in approaches that exploit laser-induced surface defects as catalytic sites, and pulsed laser ablation can induce specific types of defects on

the metal surfaces. These defects, such as lattice distortions, dislocations, twin boundaries, and coordinatively unsaturated atoms, can effectively trap electrons and reduce reaction energy barriers, thereby serving as highly active catalytic sites for the reduction of  $\text{Cu}^{2+}$  [118]. Xiang et al. [119] employed copper phosphate ( $\text{Cu}_3(\text{PO}_4)_2$ ) or nickel phosphate ( $\text{Ni}_3(\text{PO}_4)_2$ ) as a laser-active precursors to fabricate metallic patterns on ceramic substrates via direct laser activation and electroless plating. Upon laser irradiation,  $\text{Cu}_3(\text{PO}_4)_2$  was reduced to  $\text{Cu}^+$ , whereas  $\text{Ni}_3(\text{PO}_4)_2$  was converted to  $\text{NiO}$ . The resulting  $\text{Cu}^+$  or  $\text{NiO}$  adsorbed on the porous sponge-like microstructure of modified polycarbonate (PC) to form catalytic active centers that enabled selective deposition copper deposition confined to the activated regions. Meanwhile, the deposited copper layers exhibited excellent selectivity, high electrical conductivity, and strong adhesion to the substrate. Liu et al. [120] established a high-precision molecular dynamics model to simulate single-pulse picosecond laser ablation of SiC crystals. By integrating a two-temperature model (TTM) with a sub-micrometer laser focal point (on the order of one-thousandth of the spot diameter) and optimized coupling coefficients, a multi-scale correlated modeling was developed to simulate the effects of pulse overlap and laser fluence on ablation width, and the results agree well with experimental data, exhibiting a mean error below 1%. From a kinetic perspective, although traditional Pd-based catalytic systems also provide low-energy-barrier pathways, the reduction in activation energy primarily relies on the intrinsic electronic properties of Pd, such as the size and distribution density of Pd nanoparticles and their binding strength with the substrate, rather than on defect engineering. By contrast, laser-induced surface defects alter the reaction pathway via lattice strain and reduced atomic coordination, resulting in a marked lowering of the activation energy barrier.

#### 4.2. Effect of laser ablation on properties of electroless copper plating

Laser ablation is an effective method to improve the surface roughness of ceramics. Liu et al. [[121], [122], [123]] demonstrated an efficient technique for producing high-adhesion copper patterns on  $\text{Al}_2\text{O}_3$  substrate using nanosecond laser modification and ECP process. Wu et al. [124] employed lasers to fabricate diverse textures on the ceramic surface,

thereby altering the adhesion forces between the ceramic and the electroless plating coating. As depicted in Fig. 22(a), the findings from laser processing indicate that adjusting the laser-modified texture and scanning spacing can enhance the adhesion strength of the coatings. As illustrated in Fig. 22 (b-d), due to the intrinsic characteristics of different structures, the morphology and surface area of various textures exhibit distinct evolution patterns as the laser scanning spacing varies. With decreasing processing spacing, the adhesion of each coating first increases rapidly, followed by a significant decline. The adhesion of the square net texture (SNT) stripes treated by laser is 42% higher than that of the transition layer-assisted electroless plating. Xiang et al. [125] systematically investigated the influence of preparing groove structures at varying scanning speeds on the mechanical and chemical properties of AlN surface coatings. The shear strength of the AlN/Cu interface was evaluated using a universal testing machine. Following appropriate laser modification and parameter optimization, a unique porous sponge-like microstructure formed between the copper layer and the substrate, significantly enhancing the adhesion strength between them. Ermantraut et al. [126] employed laser patterning on the substrate surface, followed by electroless plating of copper, nickel, and gold in designated areas to fabricate three-dimensional ceramic interconnect devices. The resultant high-adhesion copper layer, with a bonding strength of 30N/mm<sup>2</sup>, combined with the low thermal expansion coefficient of the Al<sub>2</sub>O<sub>3</sub> substrate, ensured excellent reliability even under thermal cycling conditions.



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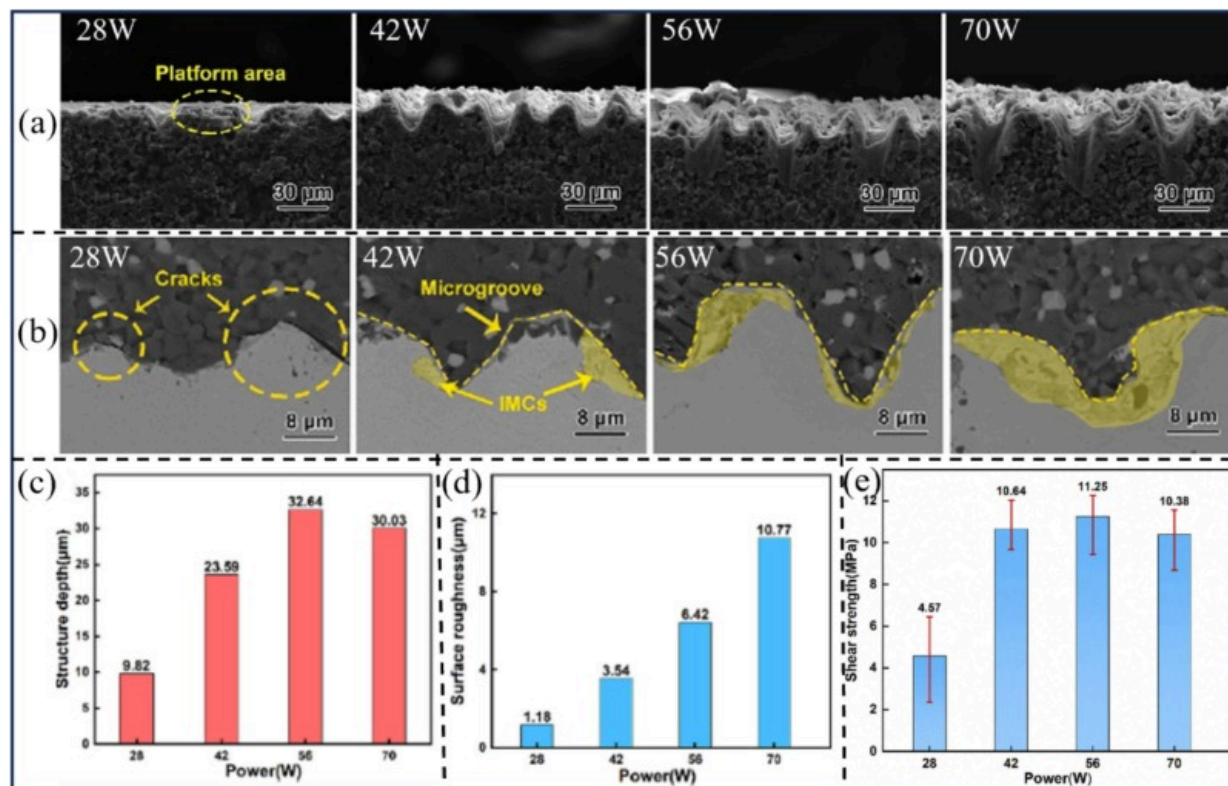
Fig. 22. Nanosecond laser processing of  $\text{Al}_2\text{O}_3$  substrate surface morphology: (a) Adhesion between chemical copper layer and ceramic matrix with different textures; (b) Surface morphologies of parallel groove texture with different scanning spacing; (c) Surface

morphologies of parallel wave texture with different scanning spacing; (d) Surface morphologies of square net texture with different scanning spacing [124].

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Liu et al. [127] employed a nanosecond laser to metallize AlN ceramics, achieving metallurgical bonding between the ceramic and copper at 620°C without additional material consumption, as illustrated in Fig. 23. By modulating laser power, they further investigated the influence of microgroove laser processing on copper coating formation. Fig. 23 (a) to (b) exhibits the microstructure of the AlN-Cu interface under varying laser power conditions. Increased laser power elevates Al content, promoting the AlN-Cu eutectic reaction and generating significant liquid phase formation. Upon cooling, a fully integrated joint interface is formed. Fig. 23 (c) to (e) respectively illustrate the average depth of the micro-grooves on the AlN ceramic surface, the surface roughness, and the average shear strength of the AlN/Cu joint under various power radiation conditions. When the power is set to 42W, the joint achieves a defect-free interface, and the thermal performance of the joint reaches its optimal level. At a power level of 56W, the average depth reaches its maximum value, achieving a maximum shear strength of 11.25MPa. However, when the power is further increased to 70W, the excessive laser flux induces splashing and remelting effects, leading to a reduction in the depth of the micro-groove and consequently affecting the bonding strength.





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Fig.23. (a) Cross-sectional SEM images of the laser metallized AlN ceramics; (b) Cross-sectional SEM images of the laser metallized AlN ceramics; (c) Structure depth of as-metallized AlN ceramic; (d) Surface roughness of as-metallized AlN ceramics; (e) Shear strength of AlN/Cu joints under various laser powers irradiation [127].

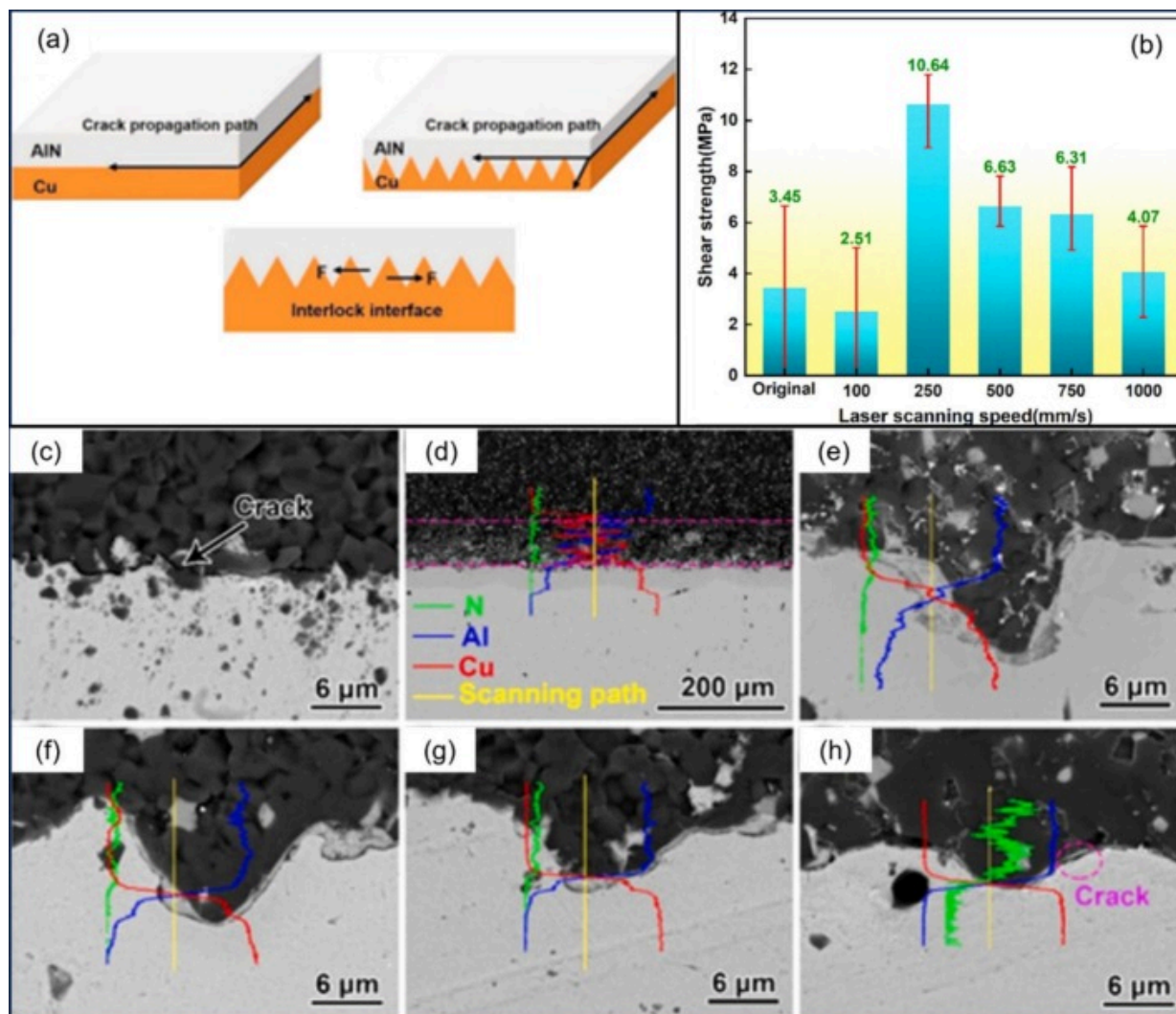
Laser ablation of surfaces is induced by either thermal effects from infrared photons or photochemical effects from ultraviolet photons. This process removes contaminants (e.g., oxides, oils) or materials from the substrate surface through the transition of solid matter to a dispersed phase. Laser processing modifies ceramic surfaces by altering their morphology, chemical composition, and microstructure. The resulting surface topography



increases the contact area between copper layers and ceramic substrates. Laser-generated high-aspect-ratio grooves create robust anchor-like interlocking structures. These structures replace traditional weak chemical bonds with high-kinetic-energy particles capable of penetrating the substrate. The embedded particles form chemical bonds within the substrate, enhancing adhesion strength between the ceramic surface and metal layers [75,128], this results in bond strength is more than twice that of the sample only after chemical coarsening [124,129].

The mechanical linkage strengthening of the interface between laser ablative ceramics and copper is shown in Fig. 24 (a). Fig. 24(a) also represents the shear strength of laser ablation of AlN ceramics at scanning speeds of (c) the original, (d) 100mm/s, (e) 250mm/s, (f) 500mm/s, (g) 750mm/s, and (h) 1000mm/s, respectively. Due to the lack of metallurgical connection at the interface, the original AlN cannot form a strong bond with copper. However, under laser irradiation, with the increase of scanning speed, the shear strength first increases and then decreases. Compared with the planar interface, the crack growth path of the grooved interlock interface changes from 2D to 3D, as illustrated in Fig. 24 (b). On the one hand, the yield of Al decreases with the increase of scanning speed, and the insufficient Al content and insufficient reaction with copper may lead to weak mechanical linkage strength and even lead to interface defects; On the other hand, with the increase of crack propagation distance, the energy required for crack propagation increases, which can improve the bonding strength of AlN ceramics and copper. With the increase of scanning speed, the depth and surface area of the interface groove decrease, which is also the reason for the weakening of mechanical interlocking strength of facial cleaning. Therefore, the interlocking interface with micro-groove structure can effectively improve the mechanical properties of the joint. In summary, the surface morphology and surface composition are helpful to improve the bonding strength of ceramic and copper after laser ablation. The change of surface chemical composition is mainly manifested as the increase of surface oxide layer, the increase of oxide will lead to the decrease of Al production, the insufficient reaction between Al and copper, the decrease of interface connection strength, and the weakening of mechanical chain effect. Compared with the

plane interface, the zigzag interface with interlocking groove joint is conducive to broadening the heat flux transfer area and enhancing the heat dissipation capacity, which is conducive to the heat transfer capacity of the joint while increasing the mechanical linkage. Adequate eutectic reaction of Al and copper is the key to ensure that AlN and copper form a complete and flawless joint. Appropriate laser processing parameters can effectively improve the structural depth and surface area of the ceramic, thereby enhancing the mechanical linkage effect in the interface. Otherwise, too slow scanning speed will cause huge thermal damage to the joint due to the pulse overlap ratio and high energy density. Therefore, it is necessary to conduct further comprehensive research on the quantitative influence of machining parameters.



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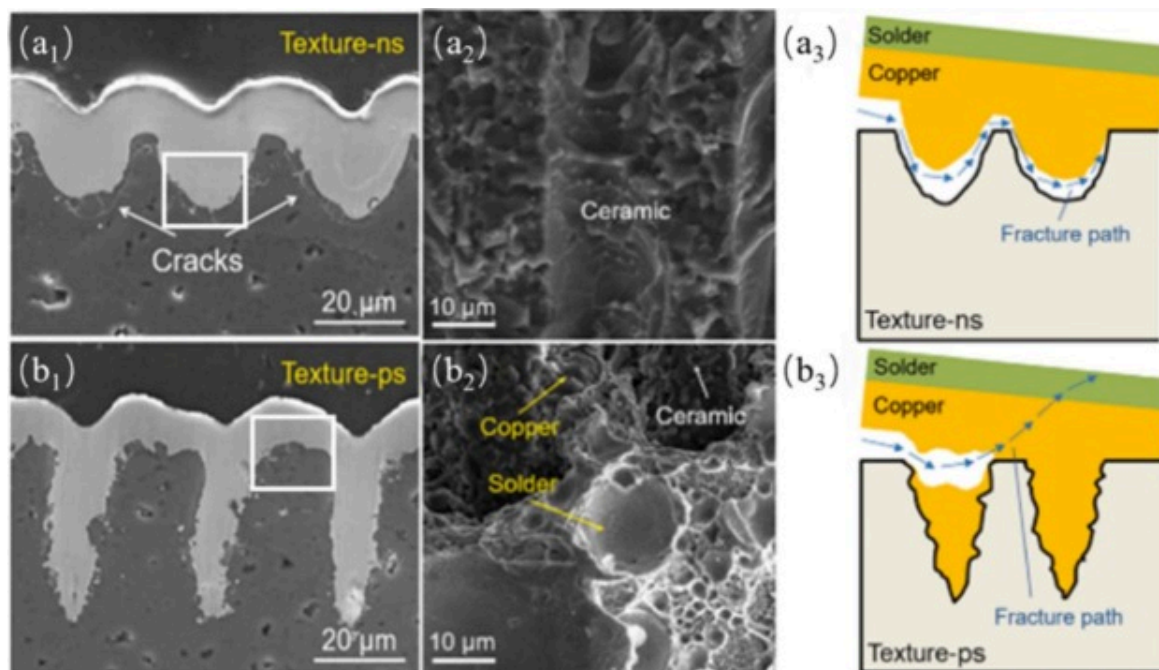
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Fig. 24. Cross-section images of the original AlN/Cu joint and the laser-irradiated-AlN/Cu with different laser scanning speeds: (a) Schematic illustration of mechanical interlock strengthening; (b) Shear strength results of AlN/Cu joints; (c) original, (d) 100mm/s, (e) 250mm/s, (f) 500mm/s, (g) 750mm/s, (h) 1000mm/s [130].

### 4.3. Failure analysis

Compared to other metallization processes, ECP has significant advantages in practical implementation, quality, environmental sustainability, and the scalability of future products. The experimental results confirm that copper ions can be directly reduced to metallic copper on laser-metallized AlN surfaces via this method, forming a uniform copper layer with a thickness of 2 to 10  $\mu\text{m}$ . Notably, batch production still encounters challenges including permeation defects, electrical shorts, open circuits, and delamination.

Ultrafast laser processing significantly enhances bonding strength compared to conventional nanosecond laser texturing techniques. This advanced approach generates high-surface-area textures on ceramic substrates while preventing thermal crack formation. Consequently, ultrafast laser-treated surfaces achieve superior adhesion to copper coatings, with bond strengths reaching 51.1 MPa [131]. Fig. 25 (a<sub>1</sub>) and (b<sub>1</sub>) present cross-sectional micrographs of chemical coating surfaces processed with nanosecond and femtosecond lasers, respectively. Both of these samples exhibit unique comb-shaped mechanical interlocking structures, in which embedded bonding mechanism dominates interfacial reinforcement at copper-ceramic junctions [132]. More explicitly, distinct fracture mechanisms emerge in picosecond versus nanosecond laser processing. Nanosecond-treated surfaces exhibit complete copper coating removal accompanied by molten material expulsion, with fracture predominantly occurring at two vulnerable sites: the copper-ceramic interface and crack-containing molten layers, as evidenced in Fig. 25 (a<sub>2</sub>) and (a<sub>3</sub>). Conversely, picosecond-processed surfaces reveal significant copper and solder retention, where fracture propagates internally through both copper and solder domains. This constitutes a mixed failure mode combining adhesive failure at interfaces and cohesive failure within copper [133], demonstrated in Fig. 25 (b<sub>2</sub>) and (b<sub>3</sub>).



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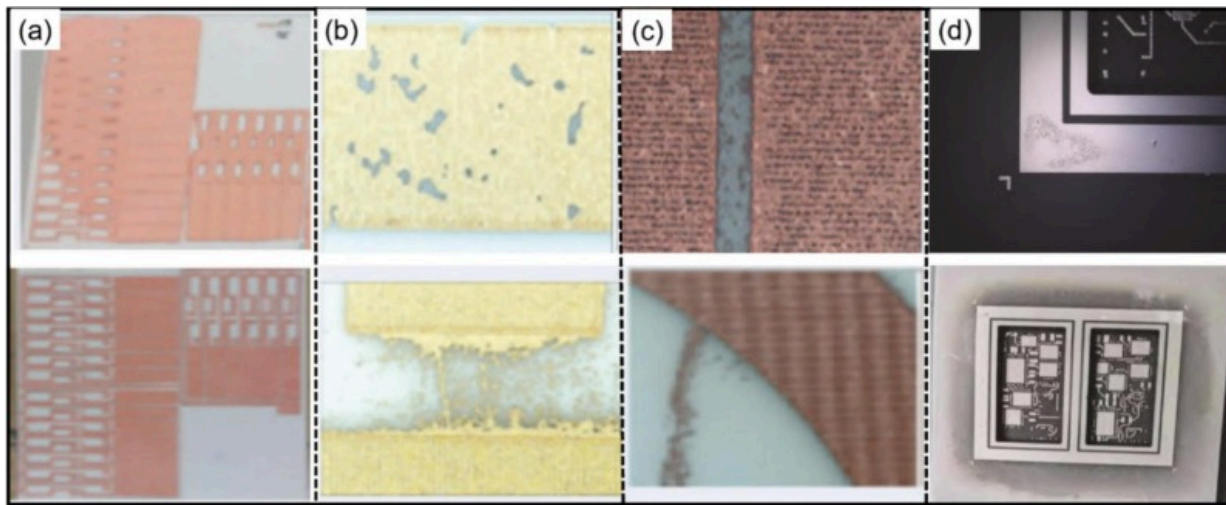
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Fig. 25. Cross-sectional views of the coatings on the textured surfaces: (a<sub>1</sub>) Texture-ns, (b<sub>1</sub>) Texture-ps; Fracture surface morphologies (a<sub>2</sub>, b<sub>2</sub>) and the schematic diagrams showing the failure mode (a<sub>3</sub>, b<sub>3</sub>) in sample Texture-ns and Texture-ps [131].

Laser-induced selective metallization modifies ceramic surface morphology and chemical composition, yet introduces challenges including process contamination and electroless plating parameter inaccuracies that degrade ceramic–metal bonding strength. During electroless copper deposition, reduced immersion time, suboptimal operating temperatures, or contaminant presence can cause plating failure, as illustrated in Fig. 26. Insufficient adhesion between the coating and the substrate, when subjected to specific thermal shock, can lead to blister formation due to stress induced by the thermal expansion of the coating. Furthermore, the presence of non-metallic, glassy phases concentrated at the metallization layer surface during substrate sintering prevents



effective bonding with the coating, resulting in bubble formation. Both phenomena impede the establishment of a robust metallurgical bond at the interface, thereby compromising bond strength. Additionally, the appearance of blackening or discoloration on metallized ceramic surfaces is attributed to the hydrolysis reaction of AlN ceramics exposed to high-temperature steam for prolonged periods. This reaction generates a mixture of poorly crystalline  $\text{AlOOH}$  and loosely adherent bayerite  $\text{Al}(\text{OH})_3$ . Implementing optimized procedures, such as thorough hot water rinsing and ensuring the cavity faces upwards during processing, effectively mitigates the formation of these black deposits [134].



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Fig. 26. Failure modes of the AlN process: (a) Plating; (b) Contamination; (c) Overlap; (d) Local area bubble image [134].

In summary, it is obvious that the laser ablation process forms an activation center on the ceramic substrate, the chemical deposition rate depends on the temperature conditions of the surface irradiation, and the distribution of laser radiation power density is not uniform with the increase of the distance of the beam focus. At the same time, the



structure and type of substrate defects are also the key factors determining the deposition rate. In the subsequent ECP process, the problem of black discoloration can be effectively avoided by changing the process, washing and cutting water in hot water after the tank is discharged, and placing the cavity facing upward when drying.

## 5. Conclusions prospects

In this paper, the impact of laser processing on the surface bonding performance of metallized ceramics and its underlying mechanisms were reviewed. Ceramics are considered as having the greatest potential in hybrid microelectronics, high-power electronics, and multi-chip modules due to the advantages of high thermal conductivity, low thermal expansion coefficient, and superior high-frequency performance. The essence of laser ablation lies in the laser heating the material surface, causing localized melting once the melting point is reached. As the temperature further rises to the boiling point, material evaporation occurs. Further increasing the incident laser power density during processing generates a plasma plume that ejects molten material, which subsequently re-solidifies on the surface. Generally speaking, laser ablation alters the surface characteristics of ceramics, primarily reflecting in surface morphology, chemical composition, and electrical properties.

After laser ablation, the micro/nano-structured surface morphology morphologies are formed on the ceramic surface, resulting in increased surface roughness. This provides a larger surface contact area for copper ions, thereby strengthening the mechanical interlocking between the ions and the ceramic substrate, which contributes to improve the adhesion of the copper coating. In addition, processing the ceramic in an inert gas atmosphere is conducive to the formation of a new metal layer on the substrate surface. The reason for this is that the surface Al content increases as oxygen availability decreases. Meanwhile, the inert environment promotes laser ablation modification and enhances the adhesion strength between copper ions and the ceramic substrate. The laser energy density is a critical process parameter in this regard. Single-pulse laser processing is laborious and time-consuming, employing a combined pulse sequence enables the

faster formation of mechanical interlocking on the ceramic surface. This approach also can reduce the formation of surface oxides, improve bond strength, minimize irregular protrusions generated at the edges of the micro-textures, and consequently enhance the quality of the deposited layer.

Although there are plenty of studies revealing the influences of laser ablation on ceramic metallization and adhesion strength, it still requires further research focusing on the intrinsic mechanisms governing laser-induced surface characteristics and bonding performance. The changes of surface morphology on laser-ablated ceramic materials are inevitably accompanied by modifications in surface chemical compositions. Therefore, it is difficult to separate the contributions of surface morphology and chemical composition on the surface wettability as well as on the adhesive bonding performance. Considering that there are many surface textures generated by laser ablation, the specific influence mechanism on the wettability and bonding performance still remain for further studies to find out the identify optimal texture parameters. Significant research gaps persist regarding the fundamental mechanisms of laser ablation for ceramic surface treatment. Firstly, the effects of the newly formed metallic and oxide layers on the ceramic-adhesive interfacial bonding strength are not yet fully elucidated. Secondly, the influence of laser ablation processes on ceramic bonding performance under varying loading conditions and environmental exposures requires further in-depth investigation. Lastly, although the interlocking micro-groove structures generated by laser ablation have been demonstrated to enhance the ceramic-metal interfacial strength, the underlying mechanisms remain insufficiently understood. Specifically, a precise quantification of the relationship between laser-induced microstructural features, such as micro-groove morphology and surface chemistry, and interfacial adhesion is needed. Addressing these knowledge gaps will provide a more robust theoretical foundation for the engineering application of laser-based ceramic surface treatment.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

No data was used for the research described in the article.

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